

07/11/25

PH4207: QIP

Ref in Nielsen & Chuang (Cambridge U.P.) → "

iii) John Preskill Lecture Notes

(<https://www.presskill.caltech.edu/PH229>)

iv) Physics of Quantum Information by Anton Zeilinger, Arther

GRADING

Mt MidSEM → 20 Marks

CT → Best 2 out of 3 → 20 Marks

Class Participation → 10 Marks

End Sem → 30 Marks

Assignments → 10/20 4-5 Assignment
classmate - 10 (opt)

50 Marks

o Classical and Quantum Computers.

o Good understanding of linear Algebra → Must Prerequisite.

o Singular Value Decomposition (SVD)

o Convex Sum (in ~~prob~~ Algebra)

o Density Matrix. → Pure Vs Mixed State.

o Quantum FT. & Discrete FT.
(QFT)

o Quantum Algorithm → Deutsch's Algorithm
Shor's Algorithm → QFT → Quantum Circuit
Grover's Algorithm

o Entanglement (~~Matrix~~ Tensor Product State)

o Bell's Inequality

o Quantum Protocol 8. → Dense Coding
→ Teleportation → Anton Zeilinger

o Quantum Optics.

o Experimental Teleportation.

o Decoherence vs Open Quantum System.

o Experimental Quantum Computation.

Common Wealth Fellowship (Sept.) } England.
Nehru Common Wealth Fellowship

o Netherlands → (Q.C)

Delft Univ.
Twente.

o Australia → Sydney, UNSW.
↳ VISA delay

o UK → Singapore.

Thesaurus

SU(2)

USA
Aus
Canada

GRE
TOEFL

1. 5 Vol. Berkley
Phy Course
2. 3 Vol. of Feynman.

Germany → I.

IMPRS → Max Planck Inst.
DARD

England

Rhodes - Oxford } October.
Gates - Cambridge

08.01.26

Introduction/Overview of the
Historical Development of the subject

[From Neilson Chung]

Theory Computer Science 1939 → Father of C.S → Alan Turing.

• Universal Turing Machine: (UTM)

Can be used to simulate
any other turning machine
that can solve a problem.

→ It completely captures what it
means to perform a task by an
algorithmic process.

↓
Give the ~~idea~~ first
abstract notion
of Programmable
computer (Math. Concepts)

★ ★ Equivalence: It means if an algorithm can be performed on
a piece of ~~hardware~~ hardware (i.e. a P.C.) then there is an
equivalent algorithm for a UTM, which perform the same task
exactly as the Algorithm running on a P.C.

↪ Also called Church-Turing
Hypothesis. thesis.

• Alonso Church

A Quantum Comp. can be a UTM ↪ Solve Classical
Solve quantum

Physical Concept of what class of algorithm can ^{be} performed on
some Physical device

↕
Rigorous mathematical concept of the UTM

Simultaneously rapid development was taken place from the Hardware
sides

Analog devices / Electronics industries / Semiconductor industries

at 1947 ATNT Bell Labs → Bardeen, Brattain, Shockley

~ discover Transistor.

Bardeen → Also discover Super conductor.

Transistor → Revolutionary chip manufacture.

Cutoff & Saturation }
○ Binary
OR
± Analog
Signal.

Giorden Moors (1975) → Moors Law. (*)

Computer Power will double with constant cost every 2 years.

Number of transistor per unit area increases → chip size reduces.

▷ As you keeps on increasing, the size → limit reach
↓
(Classical Physics Not work) + nano regime



Soln Move to a new Paradigm → Quantum Computer which uses QM as its operating sys. (of layers of QM) to perform computation.

▷ A classical QM can't simulate QM but not efficiently.

Efficient VS Inefficient Algo.

↓
Polynomial time ↓

(in size of the problem)
Solve

↓
Super polynomial or Expo in time

~~Q(NK)~~

$n \rightarrow t$

Size $\uparrow\uparrow \rightarrow$ exponentially time $\uparrow\uparrow\uparrow$

$n \rightarrow t$

$\uparrow\uparrow$
Size of the problem

Size $\uparrow\uparrow \rightarrow$ ~~that~~ linear time increase

Computational Complexity

Strong Church-Turing Thesis
(1970's)

Any Algorithm process can be simulated efficiently using a UTM.

Solovay & Strassan → Randomised algorithm → posed further challenge to the Strong Church Turing Thesis.

RSA → Password Encryption.

Factorization is a hard problem.

Can ~~not~~ determine with some probability → whether a number is prime or composite.

↓
I can

If you run few times then eventually give correct answer.

Modified Solovay & Stra

Modified Strong Church-Turing Thesis $\rightarrow$

Any Algorithm process can be simulated efficiently using a Probabilistic Turing Machine.

David Deutsch (1980) (Oxford) $\rightarrow$ Alan Turing of Quantum Comp.
 $\rightarrow$ Based on laws of Phys. $\rightarrow$ Quantum.

Efficiently simulate everything (Even Probabilistic calculation)

Gate Universal Quantum Computer.

09/01/25

Quantum Information

Feynman 1982 $\rightarrow$ Quantum Computer (Nature is inherently Quantum)

Quantum Many Body Theory. $\rightarrow$ (Classical kinda) in computer.

In Quantum Computer $\rightarrow$ Algorithm has to be Q.M.

Peter Shor (1994) $\rightarrow$ Algorithm that enables finding ~~prime~~ prime factors.

AT&T Bell's Lab.

★ ★ This is a hard problem $\rightarrow$ given the problem $\rightarrow$ difficult to solve
If soln is given $\rightarrow$ Easy to verify.

$$n = pq \quad p, q \in \text{Prime} \quad 15 = 3 \times 5$$

Given ~~soln~~ p and q we can easily find $n = pq$ but given n it is diff to find p, q.

RSA Cryptography $\rightarrow$ used (protection of password)

Number field Sieve $\rightarrow$ Best known Prime factor algo.
Required $\rightarrow T \propto \exp [C(\ln N)^{1/3} \ln(\ln N)^{2/3}]$

$$C = \left(\frac{64}{3}\right)^{1/3} \approx 9$$

a) 65 digits factors of a 130 digit Number

$\rightarrow$ Time taken to find Number $\rightarrow$ One month (Network of 100 workstations)

a) 400 digit # $\rightarrow 10^{10}$ yes

But Shor's algo factor in polynomial $O((\ln N)^3)$
(Q.M.)

a) $\rightarrow$ One month

b) $\rightarrow$ 3 years

As we can see Quantum Paradoxism.

Follow Lec

Classical bit $\rightarrow 0$ OR 1

$\downarrow$ Cutoff $\downarrow$ Saturation of a C.E. Transistor

Read

o Field Effect Transistor $\rightarrow$ use Drift instead of diffusion (unlike BJT Transistor)

o Drift vs Current vs Diffusion Current

o Semi Conductor

$$\sigma_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \cancel{\sigma_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}$$

Quantum Bit: Qubits $H \rightarrow \{ |0\rangle, |1\rangle \}$

an arbitrary state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \alpha, \beta, c \in \mathbb{C}$

$$\langle 0|0\rangle = \langle 1|1\rangle = 1, \langle 0|1\rangle = 0 \quad |\alpha|^2 + |\beta|^2 = 1$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad |\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad \langle\psi| = (\alpha^* \ \beta^*)$$

$$\langle\psi|\psi\rangle = 1$$

Examples

o TTL Logic. Read $\rightarrow$ Art of Electronics Horowitz & Hill.

o Qubit $\rightarrow$ 2 level system $\begin{matrix} |e\rangle \\ |g\rangle \end{matrix} \rightarrow$

Examples: (Physical qubits)

1. Electrons spins: $| \uparrow \rangle$ or $| \downarrow \rangle$
2. Nuclear spins:
3. A 2 level system atom

$$\begin{matrix} |1\rangle = |0\rangle = |g\rangle \\ | \downarrow \rangle = |1\rangle = |e\rangle \end{matrix}$$

o EPR $\rightarrow$ Read

1. A Cooper pair in a box (Superconducting Conduction qubits)
- J-J junction

5. Ion trap

6. Cavity QED.

7. Dopants in Silicon Quantum Dots.

8. NV centering Diamonds

10. Photon Polarization.

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$|0\rangle = \frac{1}{\sqrt{2}} (|+\rangle + |-\rangle)$$

$$|1\rangle = \frac{1}{\sqrt{2}} (|+\rangle - |-\rangle)$$

o Time Dependent Perturbation Theory.

System: 2 qubit system.
2 spin half system. : combining.

$$S_1 = \frac{1}{2}, S_2 = \frac{1}{2}$$

$$\text{Total spin} \rightarrow S_T = S_1 + S_2 = \frac{1}{2} + \frac{1}{2} = 1 \begin{cases} m_s = +1 \\ m_s = 0 \\ m_s = -1 \end{cases}$$

$$= \frac{1}{2} - \frac{1}{2} = 0 \rightarrow m_s = 0$$

$$\begin{aligned} |S_T, m_s\rangle &= \{ |1, 1\rangle, |1, 0\rangle, |1, -1\rangle, |0, 0\rangle \} \\ &= |\uparrow\uparrow\rangle \quad \downarrow \quad |\downarrow\downarrow\rangle \quad \downarrow \\ &\quad \frac{1}{\sqrt{2}}(|1, 0\rangle + |0, 1\rangle) \quad \frac{1}{\sqrt{2}}(|0, 1\rangle - |1, 0\rangle) \end{aligned}$$

Hilbert space for 2 qubit sys.

$$\{ |00\rangle, |01\rangle, |10\rangle, |11\rangle \}$$

Recap: 2 Qubit Sys $\rightarrow$ $|00\rangle, |01\rangle, |10\rangle, |11\rangle \rightarrow$ direct product.

$\dim = 2^n \rightarrow n$ qubit.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \begin{matrix} S_T = 1; m_T = 1, S_T = -1 \\ S_T = 0; m_T = 0 \end{matrix} \quad \begin{matrix} 3+1 \\ \hookrightarrow \text{direct sum} \end{matrix}$$

$\mathcal{H}_A; \mathcal{H}_B$

$\rightarrow$ Tensor Product.

A $\rightarrow$ Alice

B $\rightarrow$ Bob.

$$\mathcal{H}_A \otimes \mathcal{H}_B = |00\rangle = |0\rangle \otimes |0\rangle$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |00\rangle$$

$$|01\rangle = |0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle$$

$$|10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}; |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

These are "orthonormal" basis.

3-qubit System $\rightarrow 2^3 = 8$ member (basis).

$|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle$

$$|000\rangle = |00\rangle \otimes |0\rangle$$

HW $\rightarrow$ Do same like Previous

$$= \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$|001\rangle = |00\rangle \otimes |1\rangle = \begin{pmatrix} 1 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$|010\rangle = |01\rangle \otimes |0\rangle$$

$$= \begin{pmatrix} 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 1 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|100\rangle = |10\rangle \otimes |0\rangle$$

$$= \begin{pmatrix} 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 1 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$|011\rangle = |01\rangle \otimes |1\rangle$$

$$= \begin{pmatrix} 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 1 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$|101\rangle = |10\rangle \otimes |1\rangle$$

$$= \begin{pmatrix} 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 1 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$|110\rangle = |11\rangle \otimes |0\rangle$$

$$= \begin{pmatrix} 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 1 & \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$|111\rangle = |11\rangle \otimes |1\rangle$$

$$= \begin{pmatrix} 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 1 & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

System $\rightarrow$ an arbitrary state can be formed as.

$$|\psi\rangle = \sum_{i=0}^{2^n-1} a_i |i\rangle \quad ; \quad \sum_i |a_i|^2 = 1 \quad \langle i_j | i_k \rangle = \delta_{ij}$$

Ex. 2 qubit.

$$|\psi\rangle = a_0|00\rangle + a_1|01\rangle + a_2|10\rangle + a_3|11\rangle$$

$$|a_0|^2 + |a_1|^2 + |a_2|^2 + |a_3|^2 = 1.$$

Di Vincenzo Criteria - David P. DiVincenzo

- (i) Should be able to initialize all the states.
- (ii) Should be able to perform computation:-
(Unitary Evolution on n -qubit state.

$U_{2^n \times 2^n} \rightarrow$ Unitary matrix.

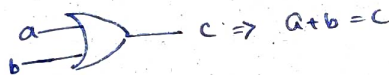
- (iii) Should be able to evolve the system long enough to avoid decoherence.

- (iv) Should be able to readout the results - Quantum Measurement.

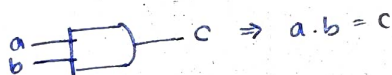
- (v) i.e. $|\psi\rangle = \sum_i a_i |i\rangle \rightarrow$ Measurement readout $\Rightarrow |i_j\rangle$ with prob. probability $|a_j|^2$.

Gates:-

(i) OR ~~GATE~~ Gate:-



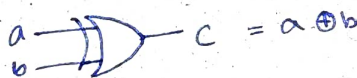
(ii) AND Gate



(iii) NOT Gate



(iv) XOR Gate



$\Rightarrow$ 2 input $\rightarrow$ one output
Irreversible gate

a	b	c
0	0	0
0	1	1
1	0	1
1	1	0

$\frac{1}{10}$

Classical computation is a reduction of Computation Phase Complexity
 $2 \rightarrow 1$. (Dim. Reduce)

Toffoli Gate:-

3 inputs a, b, c Output $a, b, c \oplus a.b$

Input			Output		
a	b	c	a	b	$c \oplus a.b$
0	0	0	0	0	0
0	1	0	0	1	0
1	0	0	1	0	0
1	1	0	1	1	1
0	0	1	0	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	0

Invertible classical gate.

Deutsch's Algo (Power of Parallelism)

function $\rightarrow f(x) \rightarrow$ examine if it is constant or ~~Blat~~ Balance.

$x = 0 \text{ OR } 1$
 $f(0) = f(1) \rightarrow$ const. & balanced $\Rightarrow f(0) \neq f(1)$

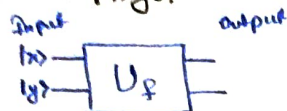
Black box



$\rightarrow$ takes 24 hours to evaluate $f(x)$ Then Need 48hr to find $f(0)$ and $f(1)$

$$\left. \begin{array}{l} f(0) \rightarrow \begin{array}{c} 0 \\ \text{OR} \\ 1 \end{array} \\ f(1) \rightarrow \begin{array}{c} 0 \\ \text{OR} \\ 1 \end{array} \end{array} \right\} \text{4 possibilities.}$$

Statement of Algo:



2 input black-box

2 qubit gate $\rightarrow$ 4D Hilbert Space

Input $|x\rangle|y\rangle \rightarrow$ Direct product
 Output $|x\rangle \otimes |y \oplus f(x)\rangle$

Digression

Hadamard Gate: H

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{Phan don't matter}$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$$

$$H|0+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1+1 \\ 1-1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

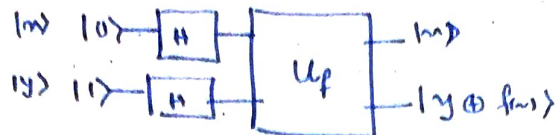
$$H|1-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1-1 \\ 1+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

$$H^2 = I \rightarrow \text{Invertible Gate.} \quad \boxed{H^{-1} = H}$$

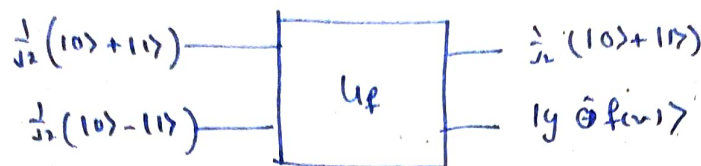
$|x\rangle = |0\rangle$

Initialization

$|y\rangle = |1\rangle$



Step-2



Output ψ

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \{|y \oplus f(x)\rangle\}$$

~~$y \oplus f$~~

$$= \frac{1}{\sqrt{2}} |0\rangle |y \oplus f(0)\rangle + \frac{1}{\sqrt{2}} |1\rangle |y \oplus f(1)\rangle$$

$$= \frac{1}{\sqrt{2}} |0\rangle \frac{1}{\sqrt{2}} (|0\rangle \oplus f(0)\rangle - |1\rangle \oplus f(0)\rangle) + \frac{1}{\sqrt{2}} |1\rangle \frac{1}{\sqrt{2}} (|0\rangle \oplus f(1)\rangle - |1\rangle \oplus f(1)\rangle)$$

If $f \rightarrow$ const.

(i) $f(0) = f(1) = 0$ let

$\frac{1}{\sqrt{2}} f$

$$\frac{1}{\sqrt{2}} |0\rangle \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) + \frac{1}{\sqrt{2}} |1\rangle \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |+\rangle |-\rangle$$

So then if we put a $\boxed{H}$ on $|x\rangle \Rightarrow |0\rangle \oplus$ The const.

(ii) $f(0) = f(1) = 1$ (let)

same

If $f(0) \neq f(1) \Rightarrow \psi = \frac{1}{\sqrt{2}} \otimes (-1)^{f(x)} (|0\rangle - |1\rangle) \times \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$

So on putting the $\boxed{H}$ on first input we can get $|1\rangle$

So if $f \Rightarrow$ const then $\boxed{H}$



if $f \Rightarrow$ balanced



$$U_f|m\rangle|y\rangle = |m\rangle|y \oplus f(m)\rangle$$

$$f(m) = 0 \text{ OR } 1$$

15/11/26 Lecture-4



$$\begin{aligned} |m\rangle|y\rangle &= \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{\sqrt{2}}|0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) + \frac{1}{\sqrt{2}}|1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \end{aligned}$$

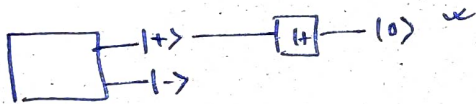
$$\begin{aligned} U_f|m\rangle|y\rangle &= \frac{1}{\sqrt{2}} U_f|0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) + \frac{1}{\sqrt{2}} U_f|1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{\sqrt{2}}|0\rangle \otimes \frac{1}{\sqrt{2}}(|0 \oplus f(0)\rangle - |1 \oplus f(0)\rangle) \\ &\quad + \frac{1}{\sqrt{2}}|1\rangle \otimes \frac{1}{\sqrt{2}}(|0 \oplus f(1)\rangle - |1 \oplus f(1)\rangle) \end{aligned}$$

Case-1 f is const $f(0) = f(1)$

Assume $f(0) = f(1) = 0$ (a)

$$\begin{aligned} &= \frac{1}{\sqrt{2}}|0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) + \frac{1}{\sqrt{2}}|1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= |+\rangle \otimes |-\rangle \end{aligned}$$

Now



If f is const
Then $|0\rangle$

(b) f is cut $f(0) = f(1) = 1$

$$\begin{aligned} &= \frac{1}{\sqrt{2}}|0\rangle \otimes \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) + \frac{1}{\sqrt{2}}|1\rangle \otimes \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) \\ &= -\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= -|+\rangle \otimes |-\rangle = |+\rangle \otimes |-\rangle \end{aligned}$$



If f is balanced
Then $|1\rangle$

Case-2 $f(0) \neq f(1)$

(a) $f(0) = 1$; $f(1) = 0$

$$\begin{aligned} &= \frac{1}{\sqrt{2}}|0\rangle \otimes \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) + \frac{1}{\sqrt{2}}|1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= -\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= -|-\rangle \otimes |-\rangle \end{aligned}$$



$$f(0) = 0; f(1) = 1$$

$$\frac{1}{\sqrt{2}} |0\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} |1\rangle \otimes \frac{1}{\sqrt{2}} (|1\rangle - |0\rangle)$$

$$= \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$= |-\rangle |-\rangle \quad \checkmark$$



$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

$$y \oplus 0 = y$$

$$f(x) = 0 \text{ or } 1 \quad (1-f(x) = 0)$$

$$(i) f(x) = 0, \quad |x\rangle |y \oplus f(x)\rangle = |x\rangle |y \oplus 0\rangle = |x\rangle |y\rangle$$

$$= \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$(ii) f(x) = 1$$

$$U_f |x\rangle |y\rangle = U_f |x\rangle \left\{ \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \right\}$$

$$U_f |x\rangle |y\rangle = |x\rangle \left\{ \frac{1}{\sqrt{2}} (|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle) \right\}$$

$$= |x\rangle \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

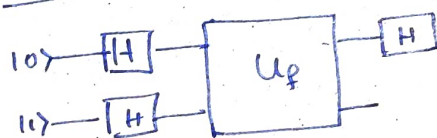
$$(ii) f(x) = 1, \quad U_f |x\rangle |y\rangle = |x\rangle \left\{ \frac{1}{\sqrt{2}} (|1\rangle - |0\rangle) \right\}$$

$$= (-1) |x\rangle \frac{1}{\sqrt{2}} \{ |0\rangle - |1\rangle \}$$

$$= (-1) |x\rangle |y\rangle$$

$$y \oplus 1 = \bar{y}$$

$$\text{So } U_f |x\rangle |y\rangle = (-1)^{f(x)} |x\rangle |y\rangle$$



$$U_f |x\rangle |y\rangle = U_f \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$= (-1)^{f(x)} \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$= \frac{1}{\sqrt{2}} U_f |0\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) + \frac{1}{\sqrt{2}} U_f |1\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$= \frac{1}{\sqrt{2}} (-1)^{f(0)} |0\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) + \frac{1}{\sqrt{2}} (-1)^{f(1)} |1\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$= \frac{1}{\sqrt{2}} ((-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \quad \text{--- (ii)}$$

3rd step

Invoke

$$f(n) = \text{count} \rightarrow f(0) = f(1) \Rightarrow (-1)^{f(0)} = (-1)^{f(1)} \xrightarrow{\text{Can-1}} \text{HA} \otimes I_B \rightarrow |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\otimes U_f(n) |y\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$= |+\rangle \otimes |-\rangle$$

↓
Apply a Hadamard Gate $\rightarrow |0\rangle$

$f(0) \neq f(1)$; $f(n)$ is not balanced $\xrightarrow{\text{Can-2}}$
 There is a diff of (-1) term

$$U_f(n) |y\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \xrightarrow{I_A \otimes I_B} |1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$= |-\rangle \otimes |-\rangle$$

↓
Apply Hadamard Gate $\rightarrow |1\rangle$ ✓

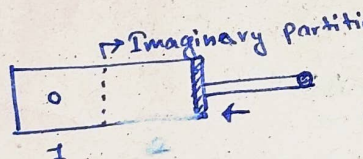
Quantum Parallelism $\xrightarrow{\uparrow}$

Deutsch, Deutsch-Jozsa Algo $\rightarrow$ Higerdim Algo.

Information is Physical $\rightarrow$ (Rolf Landauer 1961)

$a \rightarrow c = a+b$ $\left\{ \begin{array}{l} \text{Compression} \\ \text{Irreversible Reduction of} \\ \text{Complexity.} \end{array} \right.$ Computational Phase space
 $a \rightarrow c = a \cdot b$ Any Classical computation $\rightarrow$ Erases Erases $\rightarrow$ Heat Produce.

Theorem: Each time a single bit of information is erased, the amount of energy Heat dissipated into the environment is at least $k_B T \ln 2$ where T is the temperature of the surrounding Environment.


 Imaginary Partition
 Entropy = $k_B \ln 2 = S$ $\xrightarrow{\uparrow}$ Partition Number
 We don't know where the particle is between 1 and 2.
 if we change of entropy after compression $\rightarrow$ we know where the particle is $\rightarrow S_F = k_B \ln 1 = 0$

So change $\rightarrow \Delta S = k_B \ln 2$

So, Heat dissipation $\Delta Q = T \Delta S = k_B T \ln 2$

Book: "Information is Entropy"

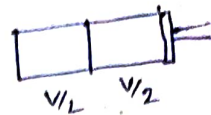
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Lecture-5

Landauer's Theorem (1961) ... check Spelling.

Dissipation $\rightarrow TAS = k_B T \ln 2$

Follow the piston example from previous class



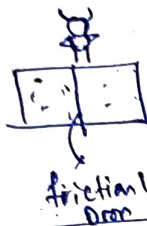
$$W = - \int_{V_i=V}^{V_f=V/2} P dV = - \int_V^{V/2} \frac{k_B T}{V} dV = - k_B T \ln \frac{V/2}{V} = k_B T \ln 2 = \Delta Q$$

Heat dissipation.

~~Proof~~

$$PV = n k_B T$$

$$n=1 \rightarrow PV = k_B T$$

Maxwell's Demon's Paradox:-

Demon allows slow moving particles ~~to~~ to filter through the orifice (opens the door) from left side to the right side and allows fast moving particles to filter through from right side to left side.

After some time



$$T_2 > T_1$$

Without doing any work the demon managed to create a temp diff.

Landauer said, A demon is a demon not God, so has finite memory. Each time he makes an observation. He stores all this in his memory. So At some point, the demon has to erase some of the data stored in his memory. In this process he does work and ~~erasure~~ erasure causes dissipation of heat.

So the demon has to do work, (~~heat dissipation~~ & heat dissipation)
So we can't do perpetual machine.

Bloch Sphere

$$| \psi \rangle = \alpha | 0 \rangle + \beta | 1 \rangle \quad ; \quad | 0 \rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} ; \quad | 1 \rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad ; \quad |\alpha|^2 + |\beta|^2 = 1$$

$\alpha, \beta \in \mathbb{C}$

$$\alpha = e^{i\varphi_1} \cos \theta/2, \quad \beta = e^{i\varphi_2} \sin \theta/2$$

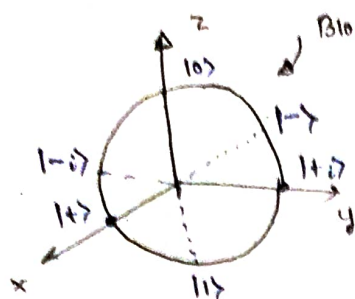
$$| \psi \rangle = e^{i\varphi_1} \cos \theta/2 | 0 \rangle + e^{i\varphi_2} \sin \theta/2 | 1 \rangle$$

$$= e^{i\varphi_1} \left(\cos \theta/2 | 0 \rangle + e^{i(\varphi_2 - \varphi_1)} \sin \theta/2 | 1 \rangle \right)$$

↓
over all phase & Define $\varphi_2 - \varphi_1 = \varphi$
we can ignore $\uparrow$

$$| \psi \rangle = \cos \theta/2 | 0 \rangle + e^{i\varphi} \sin \theta/2 | 1 \rangle$$

$$= \begin{pmatrix} \cos \theta/2 \\ e^{i\varphi} \sin \theta/2 \end{pmatrix} \quad \text{using } \textcircled{*}$$



Bloch's sphere. → unit Radius

$$(i) \boxed{\theta = 0, \varphi = 0}$$

$$|\psi\rangle = |0\rangle$$

$$(ii) \boxed{\theta = \pi, \varphi = 0}$$

$$|\psi\rangle = |1\rangle$$

$$(iii) \boxed{\theta = \pi/2, \varphi = 0}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$(iv) \boxed{\theta = \pi/2, \varphi = \pi}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$(v) \boxed{\theta = \pi/2, \varphi = \pi/2}$$

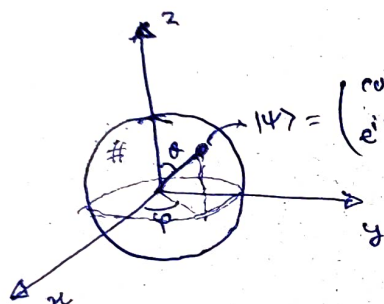
$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = |+\rangle \rightarrow \text{Notation}$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$(vi) \boxed{\theta = \pi/2, \varphi = 3\pi/2}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle) = |-\rangle \rightarrow \text{Notation}$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$



$$|\psi\rangle = \begin{pmatrix} \cos \theta/2 \\ e^{i\varphi} \sin \theta/2 \end{pmatrix}$$

★ Any point which lies on the surface of the Bloch's sphere is pure state

$|\psi\rangle$ is pure state

$$\textcircled{3} \langle \psi | \sigma_x | \psi \rangle = (\cos \theta/2 \quad e^{-i\varphi} \sin \theta/2) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \theta/2 \\ e^{i\varphi} \sin \theta/2 \end{pmatrix}$$

$$= (\cos \theta/2 \quad e^{-i\varphi} \sin \theta/2) \begin{pmatrix} e^{i\varphi} \sin \theta/2 \\ \cos \theta/2 \end{pmatrix}$$

$$= \cos \theta/2 \sin \theta/2 [e^{i\varphi} + e^{-i\varphi}]$$

$$= 2 \cos \theta/2 \sin \theta/2 \cos \varphi$$

$$\star \boxed{\langle \sigma_x \rangle_\psi = \cos \theta \cos \varphi}$$

$$\langle \psi | \sigma_z | \psi \rangle = (\cos \theta/2 \quad e^{-i\varphi} \sin \theta/2) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos \theta/2 \\ e^{i\varphi} \sin \theta/2 \end{pmatrix}$$

$$= (\cos \theta/2 \quad e^{-i\varphi} \sin \theta/2) \begin{pmatrix} \cos \theta/2 \\ -e^{i\varphi} \sin \theta/2 \end{pmatrix}$$

$$= \cos^2 \theta/2 - \sin^2 \theta/2 = \cos \theta$$

$$\star \boxed{\langle \sigma_z \rangle_\psi = \cos \theta}$$

$$\text{Calculate} \rightarrow \boxed{\langle \sigma_y \rangle_\psi = \sin \theta \sin \varphi}$$

$(\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) \rightarrow$ spherical polar ($r=1$)
ON Surface

Self Note

• Not Gate = $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\times \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \beta \\ \alpha \end{pmatrix}$

• Z-Gate = $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$ Unchanged

Note that

$Z \equiv \sigma_z$

$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\begin{pmatrix} 0 \\ 1 \end{pmatrix} = -|1\rangle$ with sign change

• Hadamard Gate (H) = $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$H^2 = I$

↳ Square Root of Not-gate

$H|0\rangle = |+\rangle$; $H|+\rangle = |0\rangle$

$H|1\rangle = |-\rangle$; $H|-\rangle = |1\rangle$

$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

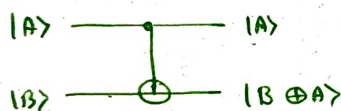
↳ Rotation of 90° by $\hat{y}$ axis $\rightarrow$ Rotation of 180° by x axis.

• Decomposing single qubit operation.

$U = e^{i\alpha} \begin{bmatrix} e^{-i\beta/2} & 0 \\ 0 & e^{i\beta/2} \end{bmatrix} \begin{bmatrix} \cos\gamma/2 & -\sin\gamma/2 \\ \sin\gamma/2 & \cos\gamma/2 \end{bmatrix} \begin{bmatrix} e^{-i\delta/2} & 0 \\ 0 & e^{i\delta/2} \end{bmatrix}$ → Euler Decomp. $Z-Y-Z$

as a product of Rotations.

• ~~Control qubit~~ CNOT gate OR Controlled NOT gate



$U_{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$\begin{matrix} y \oplus 0 = y \\ y \oplus 1 = \bar{y} \end{matrix}$

~~If control~~ It has control-qubit and target qubit

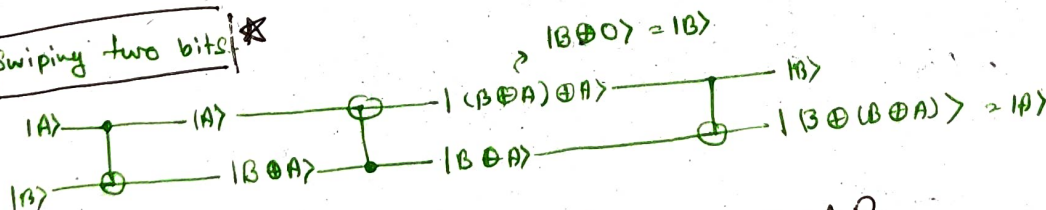
If control qubit is 0, then target qubit left alone

$U_{CNOT}|00\rangle = |00\rangle$; $U_{CNOT}|01\rangle = |01\rangle$

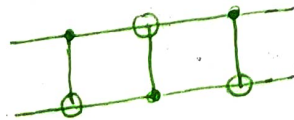
If control qubit is 1 ; the target qubit flipped

$U_{CNOT}|10\rangle = |11\rangle$; $U_{CNOT}|11\rangle = |10\rangle$

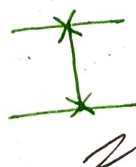
Swapping two bits *



So



=



SWAP

20/01/26

Quantum Information (CW)

$$| \psi \rangle = \cos \frac{\theta}{2} | 0 \rangle + e^{i\varphi} \sin \frac{\theta}{2} | 1 \rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\varphi} \sin \frac{\theta}{2} \end{pmatrix}$$

Revisit Euler Angle

Revisit SU(2)

Quantum gate + Rotation on Bloch Sphere.

Why do we take $\theta/2$? $| 0 \rangle$ is antipodal point of $| 1 \rangle \rightarrow$ Diametrically opposite on Bloch Sphere.How do we find antipodal point? $\theta \rightarrow \pi - \theta$; $\varphi \rightarrow \pi + \varphi$

$$| \psi(\theta, \varphi) \rangle \xrightarrow{\pi - \theta, \pi + \varphi} | \psi' \rangle = \begin{pmatrix} \sin \frac{\theta}{2} \\ -e^{i\varphi} \cos \frac{\theta}{2} \end{pmatrix}$$

$$\cos(\theta/2) \rightarrow \cos(\frac{\pi - \theta}{2})$$

$$= \sin(\theta/2)$$

$$\sin \theta/2 \rightarrow \sin(\frac{\pi - \theta}{2})$$

$$= \cos(\theta/2)$$

$$e^{i\varphi} \rightarrow e^{i(\pi + \varphi)}$$

$$= -e^{i\varphi}$$

★

$$\langle \psi' | \psi \rangle = (\sin \theta/2 - e^{i\varphi} \cos \theta/2) \begin{pmatrix} \sin \theta/2 \\ e^{i\varphi} \sin \theta/2 \end{pmatrix}$$

$$= \sin \theta/2 \cos \theta/2 - \sin \theta/2 \cos \theta/2 = 0$$

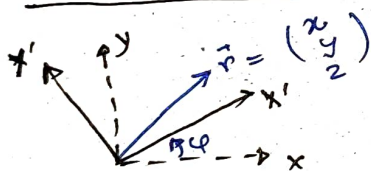
So $| \psi' \rangle \rightarrow$ Antipodal point of $| \psi \rangle$

Property of SU(2)

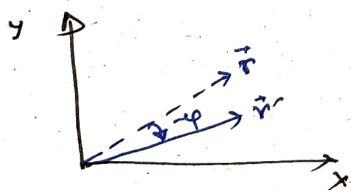
$$\text{if we do, } | \psi \rangle = \begin{pmatrix} \cos \theta \\ e^{i\varphi} \sin \theta \end{pmatrix} \xrightarrow{\pi - \theta, \pi + \varphi} \begin{pmatrix} -\cos \theta \\ e^{i\varphi} \sin \theta \end{pmatrix} = -| \psi \rangle$$

★ So it's same. $\rightarrow$ So we don't do $\pi - \theta$ instead of $\theta/2$.

Rotation Operators:-

Anti clockwise $\Rightarrow +ve$ Rotation of coordinate system $\rightarrow$ anti clockwise.
by the angle φ . $\rightarrow$ Passive Rotation.

$$(\vec{r}') = \begin{pmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = R_2(\varphi) \vec{r} \quad \text{--- (1)}$$

★ Clockwise Rotation of vector $\rightarrow$ Active Rotation

$$\vec{r}' = R_2(-\varphi) \vec{r} = \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \text{--- (2)}$$

Wave fn $\rightarrow$ Isotropy of space time the new wavefunction in the new coordinate system will be same as the old wavefunction in old coordinate system.using Active Rotation $\rightarrow$

$$\therefore \vec{r}'' = R_2^{-1}(-\varphi) \vec{r}' = R_2(\varphi) \vec{r}'$$

$$\psi'(\vec{r}') = \psi(\vec{r}) \rightarrow \text{Isotropic space}$$

$$\psi'(\vec{r}') = \psi(R_2(\varphi) \vec{r}')$$

Drop the prime
(Dummy index) --- (3)

$$\psi(\vec{r}) = \psi(R_2(\varphi) \vec{r}) \quad \text{--- (4)}$$

Consider infinitesimal Rotation: $\varphi \rightarrow \delta\varphi$
 ~~$\cos \delta\varphi = 1, \sin \delta\varphi = \delta\varphi$~~ $\cos \delta\varphi = 1, \sin \delta\varphi = \delta\varphi$

$$R_z(\delta\varphi) = \begin{pmatrix} 1 & \delta\varphi & 0 \\ -\delta\varphi & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad R_z(\delta\varphi) \vec{r} = \begin{pmatrix} 1 & \delta\varphi & 0 \\ -\delta\varphi & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$= \begin{pmatrix} x + y\delta\varphi \\ -\delta\varphi x + y \\ z \end{pmatrix} \quad \text{--- (5)}$$

from 4 and 5,

$$\begin{aligned} \psi'(\vec{r}) &= \psi(x + y\delta\varphi, y - x\delta\varphi, z) \\ &= \psi(x, y, z) + y\delta\varphi \frac{\partial \psi}{\partial x} - x\delta\varphi \frac{\partial \psi}{\partial y} \\ &= \psi(x, y, z) + \frac{i}{\hbar} \delta\varphi y p_x - \frac{i}{\hbar} \delta\varphi x p_y \\ &= \psi(x, y, z) + \frac{i}{\hbar} \delta\varphi (y p_x - x p_y) \psi \end{aligned}$$

$$\boxed{\psi'(\vec{r}) = \psi(x, y, z) - \frac{i}{\hbar} \delta\varphi \hat{L}_z} \quad \text{--- (6)}$$

Angular momentum
is the
generator of
Rotation.

For finite $\delta\varphi = \frac{\varphi}{N}$; $N \rightarrow \infty$

$$\psi'(x, y, z) = \lim_{N \rightarrow \infty} \left(1 - \frac{i}{\hbar} \frac{\varphi}{N} L_z \right)^N \psi(x, y, z)$$

$$e^{-i \frac{\varphi}{\hbar} L_z} \rightarrow \text{Rotation}$$

$$\psi' = e^{-i \frac{\varphi}{\hbar} L_z} \psi(x, y, z)$$

$$\downarrow$$

$$R_z(\varphi)$$

on
Bloch sphere

$$L \rightarrow S, \quad S_z \Rightarrow \frac{\hbar}{2} \sigma_z$$

$$R_z(\varphi) \Rightarrow e^{-i \frac{\varphi}{2} \sigma_z}$$

for an arbitrary Rotation

$$\boxed{R_{\hat{n}}(\varphi) = e^{-i \frac{\varphi}{2} \hat{n} \cdot \vec{\sigma}}} \quad \text{--- (7)}$$

$$\hat{n} = n_x \hat{i} + n_y \hat{j} + n_z \hat{k}$$

$$\boxed{R_{\hat{n}}(\varphi) = e^{-i \frac{\varphi}{2} (\hat{n} \cdot \vec{\sigma})}} \rightarrow \text{An arbitrary Rotation at an angle of } \alpha.$$

21/11/20

Rotation at arbitrary angle

☆☆

$$R_{\hat{n}}(\theta) = \exp(-i\frac{\theta}{2}\hat{n}\cdot\vec{\sigma})$$

$$R_{\hat{n}}(\theta) = \cos\frac{\theta}{2}\mathbb{I} - i\sin\frac{\theta}{2}\hat{n}\cdot\vec{\sigma}$$

$$\hat{n} = \hat{n}_x + \hat{n}_y + \hat{n}_z$$

$$\vec{\sigma} = \hat{\sigma}_x + \hat{\sigma}_y + \hat{\sigma}_z$$

$$R_{\hat{n}}(\theta) = e^{-i\hat{n}\cdot\vec{\sigma}\frac{\theta}{2}}$$

$$= \mathbb{I} - i\frac{\theta}{2}\hat{n}\cdot\vec{\sigma} + (-i)^2\frac{(\hat{n}\cdot\vec{\sigma})^2}{2!}\left(\frac{\theta}{2}\right)^2 + (-i)^3\frac{(\hat{n}\cdot\vec{\sigma})^3}{3!}\left(\frac{\theta}{2}\right)^3 + (-i)^4\frac{(\hat{n}\cdot\vec{\sigma})^4}{4!}\left(\frac{\theta}{2}\right)^4$$

$$= \left\{ \mathbb{I} + \frac{(-i)^2}{2!}(\hat{n}\cdot\vec{\sigma})^2\left(\frac{\theta}{2}\right)^2 + \frac{(-i)^4}{4!}(\hat{n}\cdot\vec{\sigma})^4\left(\frac{\theta}{2}\right)^4 \right\} + (-i)(\hat{n}\cdot\vec{\sigma})\left\{ \frac{\theta}{2} + \frac{(-i)^2}{3!}\left(\frac{\theta}{2}\right)^3(\hat{n}\cdot\vec{\sigma})^2 + \frac{(-i)^4}{5!}\left(\frac{\theta}{2}\right)^5(\hat{n}\cdot\vec{\sigma})^4 \right\}$$

Formula

$$(\vec{a}\cdot\vec{\sigma})(\vec{b}\cdot\vec{\sigma}) = (\vec{a}\cdot\vec{b})\mathbb{I} + i\vec{\sigma}\cdot(\vec{a}\times\vec{b})$$

$$\& \left[(\vec{a}\cdot\vec{\sigma})^2 = (\vec{a}\cdot\vec{a})\mathbb{I} + i\vec{\sigma}\cdot(\vec{a}\times\vec{a}) = \mathbb{I} \right]$$

$$\& R_{\hat{n}}(\theta) = \left\{ 1 - \frac{1}{2!}\left(\frac{\theta}{2}\right)^2 + \frac{1}{4!}\left(\frac{\theta}{2}\right)^4 + \dots \right\} \mathbb{I} - i(\hat{n}\cdot\vec{\sigma})\left\{ \frac{\theta}{2} - \frac{1}{3!}\left(\frac{\theta}{2}\right)^3 + \frac{1}{5!}\left(\frac{\theta}{2}\right)^5 + \dots \right\}$$

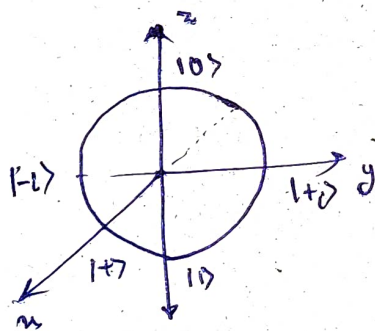
$$= \cos\frac{\theta}{2}\mathbb{I} - i(\hat{n}\cdot\vec{\sigma})\sin\frac{\theta}{2} \quad (\text{Proved})$$

* Any single Qubit Quantum Gate is a rotation around the Bloch Sphere

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H|0\rangle = |+\rangle \quad ; \quad H|1\rangle = |-\rangle$$

$|0\rangle \rightarrow |+\rangle$ Rotation about $\hat{y}$ at angle 90°



$$R_y\left(\frac{\pi}{2}\right) = e^{-i\frac{\pi}{4}\sigma_y}$$

$$= \cos\frac{\pi}{4}\mathbb{I} - i\sin\frac{\pi}{4}\sigma_y$$

$$= \frac{1}{\sqrt{2}} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right\}$$

$$= \frac{1}{\sqrt{2}} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\}$$

$$\sigma_x \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$\sigma_x \text{ in terms of } R_{\hat{x}}(\theta) \quad ; \quad R_{\hat{x}}(\theta) = \cos\frac{\theta}{2}\mathbb{I} - i\sin\frac{\theta}{2}\sigma_x$$

$$R_{\hat{x}}(\pi) = -i\sigma_x = e^{i\frac{\pi}{2}\sigma_x} = e^{-i\frac{\pi}{2}\sigma_x}$$

$$H = e^{i\frac{\pi}{2}} R_{\hat{x}}(\pi) R_y\left(\frac{\pi}{2}\right) \sim R_{\hat{x}}(\pi) R_y\left(\frac{\pi}{2}\right)$$

=

Formulae

$$(i) \boxed{G_x R_y(\theta) G_x^{-1} = R_y(-\theta)}$$

$$(ii) G_x R_z(\theta) G_x^{-1} = R_z(-\theta)$$

$$\rightarrow (iii) \boxed{R_z(\theta) G_z R_z^{-1}(\theta) = G_z} \quad *$$

$$(iv) R_x(\theta) G_y R_x^{-1}(\theta) = \cos \theta G_y + \sin \theta G_z$$

$$(v) R_y(\theta) G_x R_y^{-1}(\theta) = \cos \theta G_x + \sin \theta G_z$$

$$(vi) R_z(\theta) G_x R_z^{-1}(\theta) = \cos \theta G_x + \sin \theta G_y$$

$$(vii) R_y(\theta) G_z R_y^{-1}(\theta) = \cos \theta G_z + \sin \theta G_x$$

$$(viii) \cancel{R_y(\theta) G_z R_y^{-1}(\theta)}$$

$$* \boxed{G_i G_j = \delta_{ij} I + i \epsilon_{ijk} G_k}$$

$$[G_i, G_j] = 2i \epsilon_{ijk} G_k$$

Read
Rotation of
Goldstein

Check
H.W.

W.P.

$$R_y(\theta) G_z R_y^{-1}(\theta)$$

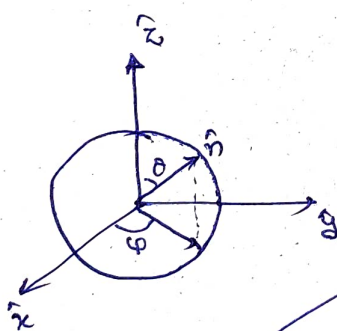
$$= \left(\cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} G_y \right) G_z \left(\cos \frac{\theta}{2} I + i \sin \frac{\theta}{2} G_y \right)$$

$$= \cos^2 \frac{\theta}{2} G_z + i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \underbrace{G_z G_y}_{-i G_x} - i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \underbrace{G_y G_z}_{i G_x} + \sin^2 \frac{\theta}{2} \underbrace{G_y G_z G_y}_{-G_z}$$

$$= \cos^2 \frac{\theta}{2} G_z + \cancel{2 \sin^2 \frac{\theta}{2}} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} G_x + \sin^2 \frac{\theta}{2} \underbrace{G_y^2}_{-I} G_z$$

$$= \cos^2 \frac{\theta}{2} G_z - \sin^2 \frac{\theta}{2} G_z + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} G_x$$

$$= \cos \theta G_z + \sin \theta G_x \quad (\text{Proved})$$



Euler Decomposition

$$\boxed{A = U A U^\dagger}$$

$$R_n(\alpha) = R_z(\varphi) R_y(\theta) R_z(\alpha) R_y^{-1}(\theta) R_z^{-1}(\varphi)$$

Rotation
anticlockwise
about z-
axis and
about
at an angle
' φ '
and take
back it $\hat{n}$.

Rotation
the vector
anticlock
wise
about y-axis
at an
angle θ that
take it to
x-z plane

Rotation
about
z-axis
by an
angle ' α '

Clockwise Rotation
of the Bloch Sphere
about y-axis by an
angle ' θ ' bring it on
the z-axis

Clockwise Rotation
of the Bloch Sphere
about z-axis by
an angle ' φ ' bring
it on the
x-z plane

$$R_Z(\varphi) (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_y) R_Z(\theta) (\cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_y) R_Z^\dagger(\varphi)$$

$$\bullet R_Z(\varphi) (\cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} \sigma_y^2)$$

$$R_Z(\varphi) (\cos^2 \frac{\theta}{2} R_Z(\theta) - i \sin \frac{\theta}{2} \sigma_y R_Z(\theta) \sigma_y -$$

$$\bullet R_Z(\varphi) R_Y(\theta) (\cos \frac{\alpha}{2} - i \sin \frac{\alpha}{2} \sigma_z) R_Y^\dagger(\theta) R_Z^\dagger(\varphi)$$

$$= R_Z(\varphi) R_Y(\theta) (\cos \frac{\alpha}{2} R_Y(\theta) R_Y^\dagger(\theta) - i \sin \frac{\alpha}{2} R_Y(\theta) \sigma_z R_Y^\dagger(\theta)) R_Z^\dagger(\varphi)$$

$$= R_Z(\varphi) [\cos \frac{\alpha}{2} - i \sin \frac{\alpha}{2} (\cos \theta \sigma_z + \sin \theta \sigma_x)] R_Z^\dagger(\varphi)$$

$$\bullet [\cos^2 \frac{\alpha}{2} R_Z(\varphi) R_Z^\dagger(\varphi) - i \sin \frac{\alpha}{2} \cos \theta R_Z(\varphi) \sigma_z R_Z^\dagger(\varphi) - i \sin \frac{\alpha}{2} \sin \theta R_Z(\varphi) \sigma_x R_Z^\dagger(\varphi)]$$

$$= [\cos^2 \frac{\alpha}{2} - i \sin \frac{\alpha}{2} \cos \theta \sigma_z - i \sin \frac{\alpha}{2} \sin \theta (\cos \varphi \sigma_x + \sin \varphi \sigma_y)]$$

$$= [\cos^2 \frac{\alpha}{2} - i \sin \frac{\alpha}{2} \cos \theta \sigma_z - i \sin \frac{\alpha}{2} \sin \theta \cos \varphi \sigma_x - i \sin \frac{\alpha}{2} \sin \theta \sin \varphi \sigma_y]$$

$$= \frac{1}{2} \cos \frac{\alpha}{2} - i \sin \frac{\alpha}{2} [\cos \theta \sigma_z + \sin \theta \cos \varphi \sigma_x + \sin \theta \sin \varphi \sigma_y]$$

$$\hat{n} \cdot \vec{\sigma}$$

$$\left\{ \begin{array}{l} \hat{n} = \sin \theta \cos \varphi \hat{i} + \sin \theta \sin \varphi \hat{j} + \cos \theta \hat{k} \\ \vec{\sigma} = \sigma_x \hat{i} + \sigma_y \hat{j} + \sigma_z \hat{k} \end{array} \right.$$

$$\text{So } R_{\hat{n}}(\alpha) = \cos \frac{\alpha}{2} \mathbb{I} - i \sin \frac{\alpha}{2} (\hat{n} \cdot \vec{\sigma})$$

Lecture 9-10

28/01/26

$$R_Y(\theta) \sigma_x R_Y^\dagger(\theta) = \cos \theta \sigma_x + \sin \theta \sigma_z$$

$$(\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_y) \sigma_x (\cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_y)$$

$$\Rightarrow \cos^2 \frac{\theta}{2} \sigma_x + i \cos \frac{\theta}{2} \sin \frac{\theta}{2} \frac{\sigma_x \sigma_y}{i \sigma_z} - i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \frac{\sigma_y \sigma_x}{-i \sigma_z} + \sin^2 \frac{\theta}{2} \frac{\sigma_y \sigma_x \sigma_y}{-\sigma_x}$$

$$\Rightarrow (\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}) \sigma_x - \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_z - \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_z$$

$$= \cos \theta \sigma_x - 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_z$$

$$\Rightarrow \boxed{\cos \theta \sigma_x - \sin \theta \sigma_z = R_Y(\theta) \sigma_x R_Y^\dagger(\theta)}$$

In general Bloch vector
can be taken as parametrize
form
(sin θ cos φ, sin θ sin φ, cos θ)
And in 2D form

Density Matrix R. K. Pathria.

$$|\psi\rangle = \sum_i c_i |\varphi_i\rangle \quad ; \quad \sum_i |c_i|^2 = 1$$

→ Arbitrary state but ~~is~~ pure state → On the surface of Bloch Sphere

$$\text{Density Matrix } \rho = |\psi\rangle\langle\psi| = \sum_i c_i |\varphi_i\rangle \sum_j c_j^* \langle\varphi_j|$$

$$\begin{aligned} &= |c_1|^2 |\varphi_1\rangle\langle\varphi_1| + |c_2|^2 |\varphi_2\rangle\langle\varphi_2| + \dots \\ &\quad + c_1 c_2^* |\varphi_1\rangle\langle\varphi_2| + c_1^* c_2 |\varphi_2\rangle\langle\varphi_1| + \dots \end{aligned}$$

$$\rho \rightarrow \begin{pmatrix} |c_1|^2 & c_1 c_2^* & \dots & c_1 c_n^* \\ c_2 c_1^* & |c_2|^2 & \dots & c_2 c_n^* \\ \vdots & \vdots & \ddots & \vdots \\ c_n c_1^* & c_n c_2^* & \dots & |c_n|^2 \end{pmatrix} \begin{matrix} \langle\varphi_1| \\ \langle\varphi_2| \\ \vdots \\ \langle\varphi_n| \end{matrix}$$

diag elements →

$$|c_1|^2, |c_2|^2, |c_3|^2, \dots, |c_n|^2$$

Diagonal elements are the Populations

$$\text{Sum of the diagonal element } \sum_i |c_i|^2 = 1 \quad ; \quad \text{Tr}(\rho) = 1$$

Eq. $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad |\alpha|^2 + |\beta|^2 = 1$

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix} = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \beta\alpha^* & |\beta|^2 \end{pmatrix} \quad \text{--- (i)}$$

$$\text{Tr}(\rho) = |\alpha|^2 + |\beta|^2 = 1$$

Population is $\frac{|\alpha|^2}{|\alpha|^2 + |\beta|^2} = |\alpha|^2$

→ Off diagonal elements are coherences

Block State

$$|\psi\rangle = \cos\frac{\theta}{2} |0\rangle + e^{i\varphi} \sin\frac{\theta}{2} |1\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\varphi} \sin\frac{\theta}{2} \end{pmatrix} ; \quad \langle\psi| = (\cos\frac{\theta}{2} \quad e^{i\varphi} \sin\frac{\theta}{2})$$

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} \cos^2\frac{\theta}{2} & e^{i\varphi} \sin\frac{\theta}{2} \cos\frac{\theta}{2} \\ e^{-i\varphi} \sin\frac{\theta}{2} \cos\frac{\theta}{2} & \sin^2\frac{\theta}{2} \end{pmatrix}$$

$$\begin{aligned} &= \begin{pmatrix} \frac{1 + \langle G_z \rangle}{2} & \frac{\langle G_x \rangle - i \langle G_y \rangle}{2} \\ \frac{\langle G_x \rangle + i \langle G_y \rangle}{2} & \frac{1 - \langle G_z \rangle}{2} \end{pmatrix} \quad \text{--- (ii)} \end{aligned}$$

$$\cos^2\frac{\theta}{2} = \frac{1 + \cos\theta}{2} = \frac{1 + \langle G_z \rangle}{2}$$

$$\sin^2\frac{\theta}{2} = \frac{1 - \cos\theta}{2} = \frac{1 - \langle G_z \rangle}{2}$$

$$e^{-i\varphi} \sin\frac{\theta}{2} \cos\frac{\theta}{2} = \cos\theta \frac{\cos\varphi - i \sin\varphi}{2} 2 \sin\frac{\theta}{2} \cos\frac{\theta}{2}$$

$$= \frac{(\cos\varphi - i \sin\varphi) \times \sin\theta}{2}$$

$$= \frac{\sin\theta \cos\varphi - i \sin\theta \sin\varphi}{2}$$

$$= \frac{\langle G_x \rangle - i \langle G_y \rangle}{2}$$

Mixed State

Ensembles of spins (10 spins)

$\uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow$

$$n_{\uparrow} = 10 \begin{cases} n_{\uparrow} = 6 \\ n_{\downarrow} = 4 \end{cases}$$

up spin = $|\uparrow\rangle = |0\rangle$

down spin = $|\downarrow\rangle = |1\rangle$

$$\rho = \frac{n_{\uparrow}}{n_{\uparrow} + n_{\downarrow}} |\uparrow\rangle\langle\uparrow| + \frac{n_{\downarrow}}{n_{\uparrow} + n_{\downarrow}} |\downarrow\rangle\langle\downarrow|$$

$$= \frac{6}{10} |\uparrow\rangle\langle\uparrow| + \frac{4}{10} |\downarrow\rangle\langle\downarrow|$$

$$\rho = \frac{e^{-\beta H}}{\text{Tr}(e^{-\beta H})} \rightarrow \text{Thermal density matrix.}$$

$$\rho = \begin{pmatrix} \langle 0| & \langle 1| \\ 0.6 & 0 \\ 0 & 0.4 \end{pmatrix} \begin{matrix} |0\rangle \\ |1\rangle \end{matrix}$$

↓
For mixed state
no information about
coherences.

↓
lost any information
related to phases.

Coherences → Precise Relate Phase.

Properties of density matrices

- (i) Trace is unity ; $\text{Tr}(\rho) = 1$
- (ii) For pure state, ρ is idempotent ; $\rho^2 = \rho$
- (iii) For Mixed States $\text{Tr}(\rho^2) < 1$

$$\rho^2 \neq \rho$$

(iv) ρ is Hermitian $\rho = \rho^\dagger$

(v) The sum of the eigenvalues of the density matrix add up to '1'

$$\sum_i \lambda_i = 1$$

For pure state, $\lambda_i = 1$; $\lambda_{i \neq j} = 0$

All eigen values are non negative

$$\langle \hat{O} \rangle = \text{Tr}(\rho \hat{O})$$

(vi) Pure state :- $|\psi\rangle$

$$|\psi\rangle = \sum_i c_i |\varphi_i\rangle ; \sum_i |c_i|^2 = 1$$

$$\langle \hat{O} \rangle = \langle \psi | \hat{O} | \psi \rangle$$

$$\text{Tr}(\rho \hat{O}) = \text{Tr}(\hat{O} \rho) = \text{Tr}(\hat{O} |\psi\rangle\langle\psi|) = \sum_i \langle \varphi_i | \hat{O} | \psi \rangle \langle \psi | \varphi_i \rangle$$

$$\text{Tr}(\rho \hat{O}) = \langle \hat{O} \rangle$$

$$= \sum_{ij} \langle \varphi_i | \hat{O} c_j |\varphi_j\rangle \sum_k c_k^* \langle \varphi_k | \varphi_i \rangle$$

$$= \sum_{ij} |c_j|^2 \langle \varphi_i | \hat{O} | \varphi_j \rangle \langle \varphi_k | \varphi_i \rangle$$

$$= \sum_{ij} c_j c_i^* \langle \varphi_i | \hat{O} | \varphi_j \rangle = \langle \hat{O} \rangle_{\psi} \text{ Proved}$$

$$= \sum_{ij} c_j c_i^* \langle \varphi_i | \hat{O} | \varphi_j \rangle = \langle \hat{O} \rangle_{\psi} \text{ Proved}$$

▷ Mixed State: Probability $\sum_j |\psi_j|^2$

$$\langle O \rangle = \sum_i P_i \langle \psi_i | O | \psi_i \rangle$$

$$\rho = \sum_i P_i |\psi_i\rangle \langle \psi_i|$$

$$= \sum_{i,j,k} P_i \underbrace{\langle \psi_i | \psi_k \rangle}_{\delta_{ik}} \underbrace{\langle \psi_k | O | \psi_j \rangle}_{O_{kj}} \underbrace{\langle \psi_j | \psi_i \rangle}_{\delta_{ji}}$$

$$= \sum_{j,k} P_i \langle \psi_j | \psi_i \rangle \langle \psi_i | \psi_k \rangle O_{kj}$$

$$= \sum_{i,k} \langle \psi_j | \underbrace{\sum_i P_i |\psi_i\rangle \langle \psi_i|}_{\rho} | \psi_k \rangle O_{kj}$$

$$= \sum_{j,k} \langle \psi_j | \rho | \psi_k \rangle O_{kj}$$

$$= \sum_{j,k} \rho_{jk} O_{kj} = \sum_j (\rho O)_j = \text{Tr}(\rho O)$$

check

~~Ascroft~~ → Magnetism → 3 chap

i) $\sigma_x R_y(\theta) \sigma_x^{-1}$

$$= \sigma_x (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_y) \sigma_x$$

$$= \cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_x \sigma_y \sigma_x$$

$$= \cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_y$$

$$= R_y(-\theta)$$

$$\sigma_x^2 = \mathbb{I}$$

$$\sigma_x \sigma_y = i \sigma_z$$

$$i \sigma_z \sigma_x = i(i \sigma_y)$$

$$= -\sigma_y$$

ii) $\sigma_x R_z(\theta) \sigma_x^{-1}$

$$\sigma_x (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_z) \sigma_x$$

$$= \cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_x \sigma_z \sigma_x$$

$$= \cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_z$$

$$= R_z(-\theta)$$

$$\sigma_z \sigma_x = i \sigma_y$$

$$i \sigma_y \sigma_x = i(i \sigma_z)$$

$$= -\sigma_z$$

iii) $R_z(\theta) \sigma_z R_z^\dagger(\theta) = (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_z) \sigma_z (\cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_z)$

$$= (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_z) (\cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_z)$$

$$= (\cos^2 \frac{\theta}{2} \mathbb{I} + i \sin + i \cos \frac{\theta}{2} \sin \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_z + i^2 \sin^2 \frac{\theta}{2} \sigma_z)$$

$$= \cos^2 \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2}) - i \sin \frac{\theta}{2} (\cos \frac{\theta}{2} - i \sin \frac{\theta}{2}) \sigma_z$$

$$= \cos \frac{\theta}{2} e^{i\theta/2} - i \sin \frac{\theta}{2} e^{-i\theta/2} \sigma_z$$

$$\begin{pmatrix} \cos \frac{\theta}{2} e^{i\theta/2} & 0 \\ 0 & \cos \frac{\theta}{2} e^{i\theta/2} \end{pmatrix} - i \sin \frac{\theta}{2} \begin{pmatrix} -i \sin \frac{\theta}{2} e^{-i\theta/2} & 0 \\ 0 & +i \sin \frac{\theta}{2} e^{-i\theta/2} \end{pmatrix}$$

$$(iii) R_z(\theta) \sigma_z R_z^\dagger(\theta) = R_z(\theta) \sigma_z R_z(-\theta)$$

$$= (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_z) \sigma_z (\cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_z)$$

$$= (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_z) (\cos \frac{\theta}{2} \sigma_z + i \sin \frac{\theta}{2} \sigma_z^2)$$

$$= (\cos^2 \frac{\theta}{2} \sigma_z + \sin^2 \frac{\theta}{2} \sigma_z - i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_z^2 + i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_z^2)$$

$$= \sigma_z$$

$$\text{So } \boxed{R_z(\theta) \sigma_z R_z^\dagger(\theta) = \sigma_z}$$

$$(iv) R_x(\theta) \sigma_y R_x^\dagger(\theta) = R_x(\theta) \sigma_y R_x(-\theta)$$

$$= (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_x) \sigma_y (\cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_x)$$

$$= (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_x) (\cos \frac{\theta}{2} \sigma_y + i \sin \frac{\theta}{2} \sigma_y \sigma_x)$$

$$= \cos^2 \frac{\theta}{2} \sigma_y + \sin^2 \frac{\theta}{2} \sigma_z - i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \underbrace{\sigma_x \sigma_y}_{i \sigma_z} + \sin^2 \frac{\theta}{2} \underbrace{\sigma_x \sigma_z}_{-i \sigma_y}$$

$$= \cos^2 \frac{\theta}{2} \sigma_y - \sin^2 \frac{\theta}{2} \sigma_y + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_z$$

$$= \cos \theta \sigma_y + \sin \theta \sigma_z$$

$$\text{Ry } \boxed{R_x(\theta) \sigma_y R_x^\dagger(\theta) = \cos \theta \sigma_y + \sin \theta \sigma_z}$$

$$(v) R_y(\theta) \sigma_x R_y^\dagger(\theta) = R_y(\theta) \sigma_x R_y(-\theta)$$

$$= (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_y) \sigma_x (\cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_y)$$

$$= (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_y) (\cos \frac{\theta}{2} \sigma_x + i \sin \frac{\theta}{2} \underbrace{\sigma_x \sigma_y}_{i \sigma_z})$$

$$= \cos^2 \frac{\theta}{2} \sigma_x - \sin^2 \frac{\theta}{2} \sigma_z - i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \underbrace{\sigma_y \sigma_x}_{-i \sigma_z} + i \sin^2 \frac{\theta}{2} \underbrace{\sigma_y \sigma_z}_{i \sigma_x}$$

$$= (\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}) \sigma_x - 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_z$$

$$= \cos \theta \sigma_x - \sin \theta \sigma_z$$

$$\boxed{R_y(\theta) \sigma_x R_y^\dagger(\theta) = \cos \theta \sigma_x - \sin \theta \sigma_z}$$

$$(vi) R_z(\theta) \sigma_x R_z^\dagger(\theta) = R_z(\theta) \sigma_x R_z(-\theta)$$

$$= (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_z) \sigma_x (\cos \frac{\theta}{2} \mathbb{I} + i \sin \frac{\theta}{2} \sigma_z)$$

$$= (\cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \sigma_z) (\cos \frac{\theta}{2} \sigma_x + i \sin \frac{\theta}{2} \underbrace{\sigma_x \sigma_z}_{-i \sigma_y})$$

$$= (\cos^2 \frac{\theta}{2} \sigma_x + \sin^2 \frac{\theta}{2} \sigma_x - i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_z \sigma_x + i \sin^2 \frac{\theta}{2} \sigma_z \sigma_x)$$

$$= \cos^2 \frac{\theta}{2} \sigma_x + \sin^2 \frac{\theta}{2} \sigma_x - i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \underbrace{\sigma_z \sigma_x}_{i \sigma_y} - i \sin^2 \frac{\theta}{2} \underbrace{\sigma_z \sigma_x}_{-i \sigma_y}$$

$$= (\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}) \sigma_x + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \sigma_y$$

$$= \cos \theta \sigma_x + \sin \theta \sigma_y$$

$$\text{Rz } \boxed{R_z(\theta) \sigma_x R_z^\dagger(\theta) = \cos \theta \sigma_x + \sin \theta \sigma_y}$$

Mixed State

$$\langle \hat{A} \rangle = \sum_k p_k \langle \psi_k | \hat{A} | \psi_k \rangle \quad p_k \Rightarrow \text{Probability of } |\psi_k\rangle \text{ state in the ensemble}$$

$$\hat{\rho} = \sum_{k, \alpha} p_k \langle \psi_k | \hat{A} | \alpha \rangle \langle \alpha | \psi_k \rangle$$

$$= \sum_{\alpha} p \langle \alpha | \left(\sum_k p_k |\psi_k\rangle \langle \psi_k| \right) \hat{A} | \alpha \rangle \quad \hat{\rho} = \sum_k p_k |\psi_k\rangle \langle \psi_k|$$

$$= \sum_{\alpha} \langle \alpha | \hat{\rho} \hat{A} | \alpha \rangle = \text{Tr}(\hat{\rho} \hat{A}) = \text{Tr}(\hat{A} \hat{\rho})$$

$$\text{So } \langle \hat{A} \rangle = \text{Tr}(\hat{A} \hat{\rho}) = \text{Tr}(\hat{\rho} \hat{A})$$

Examples of density matrix in Statistical mech.

(a) An electron in a magnetic field.

Assume applied magnetic field $\vec{H}$ then the Hamiltonian, $\hat{H} = -\gamma_e \vec{S} \cdot \vec{B}$

$$\vec{H} = \frac{\vec{B}}{\mu_0} \Rightarrow \vec{B} = \mu_0 \vec{H} \quad \gamma_e = \frac{g_e \mu_B}{\hbar} = -\frac{2e\hbar}{2m\hbar} = -\frac{e}{m}$$

$$\hat{H} = -\left(-\frac{e}{m}\right) \mu_0 (\vec{\sigma} \cdot \vec{H})$$

$$\hat{H} = \frac{e}{m} \mu_0 (\vec{\sigma} \cdot \vec{H}) \quad \hat{H} = \frac{e}{m} \mu_0 (\vec{\sigma} \cdot \vec{H}) = \frac{e}{m} \mu_0 H \hat{\sigma}_z$$

$$\hat{\rho} = \frac{e^{-\beta \hat{H}}}{\text{Tr}(e^{-\beta \hat{H}})} = \frac{e^{-\beta \frac{e}{m} \mu_0 H \hat{\sigma}_z}}{\text{Tr}(e^{-\beta \frac{e}{m} \mu_0 H \hat{\sigma}_z})} = \left(\frac{1}{e^{-\beta \frac{e}{m} \mu_0 H} + e^{+\beta \frac{e}{m} \mu_0 H}} \right) \begin{pmatrix} e^{\beta \frac{e}{m} \mu_0 H} & 0 \\ 0 & e^{-\beta \frac{e}{m} \mu_0 H} \end{pmatrix}$$

$$= \frac{1}{e^{-\beta \frac{e}{m} \mu_0 H} + e^{+\beta \frac{e}{m} \mu_0 H}} \begin{pmatrix} e^{\beta \frac{e}{m} \mu_0 H} & 0 \\ 0 & e^{-\beta \frac{e}{m} \mu_0 H} \end{pmatrix}$$

$$\langle \hat{\sigma}_z \rangle = \text{Tr}(\hat{\rho} \hat{\sigma}_z)$$

$$= \frac{1}{e^{-\beta \frac{e}{m} \mu_0 H} + e^{+\beta \frac{e}{m} \mu_0 H}} \begin{pmatrix} e^{-\beta \frac{e}{m} \mu_0 H} & 0 \\ 0 & e^{+\beta \frac{e}{m} \mu_0 H} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \frac{1}{e^{-\beta \frac{e}{m} \mu_0 H} + e^{+\beta \frac{e}{m} \mu_0 H}} \begin{pmatrix} e^{-\beta \frac{e}{m} \mu_0 H} & 0 \\ 0 & -e^{+\beta \frac{e}{m} \mu_0 H} \end{pmatrix}$$

$$= \frac{e^{-\beta \frac{e}{m} \mu_0 H} - e^{+\beta \frac{e}{m} \mu_0 H}}{e^{-\beta \frac{e}{m} \mu_0 H} + e^{+\beta \frac{e}{m} \mu_0 H}} = -\tanh\left(\beta \frac{e}{m} \mu_0 H\right)$$

29/01/25

Density Matrix

Prop III - There are 3 independent parameters which quantify the density matrix (like $SU(2)$) of qubit.

$$\rho = \begin{pmatrix} a+ib & c+id \\ e+if & g+ih \end{pmatrix}$$

Hermitian $\rightarrow \rho = \rho^\dagger$, $b=h=0$, $d=-f$, $c=e$

$$\rho = \begin{pmatrix} a & c+id \\ c-id & g \end{pmatrix}$$

$$\text{Tr}(\rho) = 1 \Rightarrow a+g=1$$

$$\rho = \begin{pmatrix} a & c+id \\ c-id & 1-a \end{pmatrix}$$

$c \rightarrow b$; $d \rightarrow e$

$$\rho = \begin{pmatrix} a & b+ic \\ b-ic & 1-a \end{pmatrix}$$

$$a = \frac{1+\alpha_3}{2}; \quad b = \frac{\alpha_1}{2}; \quad c = \frac{\alpha_2 - i\alpha_2}{2}$$

$$1-a = \frac{1-\alpha_3}{2}$$

$$\rho = \begin{pmatrix} \frac{1+\alpha_3}{2} & \frac{\alpha_1 - i\alpha_2}{2} \\ \frac{\alpha_1 + i\alpha_2}{2} & \frac{1-\alpha_3}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1+\alpha_3 & \alpha_1 - i\alpha_2 \\ \alpha_1 + i\alpha_2 & 1-\alpha_3 \end{pmatrix} = \frac{1}{2} (\mathbb{I} + \vec{\alpha} \cdot \vec{\sigma})$$

$$\boxed{\rho = \frac{1}{2} (\mathbb{I} + \vec{\alpha} \cdot \vec{\sigma})}$$

Bloch sphere representation of density matrix.

$$\frac{1}{2} \begin{pmatrix} 1+\alpha_z & \alpha_x + i\alpha_y \\ \alpha_x - i\alpha_y & 1-\alpha_z \end{pmatrix} \leftarrow$$

★ ★

$$\alpha_i = \text{Tr}(\rho \sigma_i)$$

$$\begin{aligned} \alpha_i &= \text{Tr}(\rho \sigma_i) = \text{Tr} \left\{ \frac{1}{2} (\mathbb{I} + \vec{\alpha} \cdot \vec{\sigma}) \sigma_i \right\} \\ &= \text{Tr} \left\{ \frac{1}{2} (\sigma_i + \alpha_1 \sigma_1 \sigma_i + \alpha_2 \sigma_2 \sigma_i + \alpha_3 \sigma_3 \sigma_i) \right\} \\ &= \text{Tr} \left\{ \frac{1}{2} (\sigma_i + \alpha_1 \delta_{1i} + \alpha_2 (-i\epsilon_{2i}) + \alpha_3 \delta_{3i}) \right\} \\ &= \frac{1}{2} (2\alpha_i) = \alpha_i \end{aligned}$$

$$\alpha_3 = \text{Tr}(\rho \sigma_3)$$

$$\begin{aligned} &= \text{Tr} \left\{ \frac{1}{2} (\sigma_3 + \alpha_1 \sigma_1 \sigma_3 + \alpha_2 \sigma_2 \sigma_3 + \alpha_3 \sigma_3 \sigma_3) \right\} \\ &= \frac{1}{2} (2\alpha_3) = \alpha_3 \end{aligned}$$

Corollary:- If it is a pure state,

$$\vec{\alpha} = \hat{n} = \hat{i} \sin\theta \cos\varphi + \hat{j} \sin\theta \sin\varphi + \hat{k} \cos\theta$$

$$\hat{n} = \hat{i} \langle \sigma_x \rangle + \hat{j} \langle \sigma_y \rangle + \hat{k} \langle \sigma_z \rangle$$

$$\alpha_i = \text{Tr}(\rho \sigma_i) = \langle \sigma_i \rangle$$

Bloch Vector.

True for any state (mixed or pure state)

$\rho(\vec{\alpha}) = \frac{1}{2} (\mathbb{I} + \vec{\alpha} \cdot \vec{\sigma})$ → Density vector in terms of Bloch Vector.

Property - VIII

$$\rho = \frac{1}{2}(\mathbb{I} + \vec{\alpha} \cdot \vec{\sigma}) \quad ; \quad \text{The eigenvalues, } \lambda_1 = \frac{1+|\vec{\alpha}|}{2}, \lambda_2 = \frac{1-|\vec{\alpha}|}{2}$$

$$\det\left(\frac{1}{2}(\mathbb{I} + \vec{\alpha} \cdot \vec{\sigma}) - \lambda \mathbb{I}\right)$$

$$\det \begin{vmatrix} \frac{1+\alpha_3}{2} - \lambda & \frac{\alpha_1 - i\alpha_2}{2} \\ \frac{\alpha_1 + i\alpha_2}{2} & \frac{1-\alpha_3}{2} - \lambda \end{vmatrix} = 0$$

$$\Rightarrow \left(\left(\frac{1+\alpha_3}{2} - \lambda\right)\right)^2 = \frac{\alpha_1^2 + \alpha_2^2}{4}$$

$$\Rightarrow \left(\frac{1+\alpha_3}{2} - \lambda\right)\left(\frac{1-\alpha_3}{2} - \lambda\right) = \frac{\alpha_1^2 + \alpha_2^2}{4}$$

$$\Rightarrow \left(\frac{1+\alpha_3}{2}\right)\left(\frac{1-\alpha_3}{2}\right) - \lambda\left(\frac{1-\alpha_3}{2} + \frac{1+\alpha_3}{2}\right) + \lambda^2 = \frac{\alpha_1^2 + \alpha_2^2}{4}$$

$$\Rightarrow \frac{1-\alpha_3^2}{4} - \lambda + \lambda^2 = \frac{\alpha_1^2}{4} + \frac{\alpha_2^2}{4}$$

$$\Rightarrow \lambda^2 - \lambda + \frac{(1-\alpha_3^2 - \alpha_1^2 - \alpha_2^2)}{4} = 0$$

$$\Rightarrow \lambda^2 - 2\lambda \times \frac{1}{2} + \frac{1}{4} = \frac{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}{4}$$

$$\Rightarrow \left(\lambda - \frac{1}{2}\right)^2 = \frac{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}{4}$$

$$\Rightarrow \lambda = \frac{1 \pm \sqrt{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}}{2} + \frac{1}{2}$$

$$\boxed{\lambda = \frac{1 \pm |\vec{\alpha}|}{2}}$$

→ Eigen value of $\rho(\vec{\alpha}) = \frac{1}{2}(\mathbb{I} + \vec{\alpha} \cdot \vec{\sigma})$

For a pure state, $\vec{\alpha}$ lies on the Bloch Sphere. → $|\vec{\alpha}| = 1$ - for pure state

Then $\boxed{\lambda_1 = 1, \lambda_2 = 0}$ → for pure state.

★

For maximally mixed states → $\boxed{\lambda_1 = \lambda_2 = 1/2}$

↓

$$|\vec{\alpha}| = 0$$

↓

(point lies at the center)
of the Bloch sphere

$$\rho = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



Maximally mixed state

1. For pure state $\rightarrow$ lies on the Bloch sphere

2. For maximally mixed state $\rightarrow$ lies ~~on the~~ at the center.

Quantum Computation can be done very very well for states on the surface of Bloch sphere as we need coherence, then, for maximally mixed state it is worst.

$\triangleright$ For partially mixed state, $|\alpha| < 1$, then, $\vec{\alpha}$ lies ~~is the~~ inside the Bloch sphere. (states)

$\triangleright |\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle = e^{-\frac{iHt}{\hbar}} |\psi(0)\rangle$

Note that $\hat{U}^\dagger \hat{A} \hat{U} \rightarrow \hat{A}'$

$\langle \hat{A} \rangle = \langle \psi(t) | \hat{A} | \psi(t) \rangle = \langle \psi(0) | \hat{U}^\dagger \hat{A} \hat{U} | \psi(0) \rangle = \langle \psi(0) | \hat{A}' | \psi(0) \rangle$

$\hat{A} \rightarrow \hat{U}^\dagger \hat{A} \hat{U}$

$\frac{d\langle \hat{A} \rangle}{dt} = \frac{d\hat{U}^\dagger \hat{A} \hat{U}}{dt} + \frac{d\hat{A}}{dt}$ (as $\frac{\partial \hat{A}}{\partial t} = 0$ No explicit time dependence)

$= \frac{i}{\hbar} \hat{U}^\dagger \hat{H} \hat{A} \hat{U} - \frac{i}{\hbar} \hat{U}^\dagger \hat{A} \hat{H} \hat{U}$

$\frac{d\langle \hat{A} \rangle}{dt} = \frac{i}{\hbar} \hat{U}^\dagger \hat{H} \hat{U} \hat{A}' - \frac{i}{\hbar} \hat{U}^\dagger \hat{A}' \hat{H} \hat{U}$

$= \frac{i}{\hbar} \hat{A}' \hat{H} - \frac{i}{\hbar} \hat{H} \hat{A}'$

$\frac{d\langle \hat{A} \rangle}{dt} = -\frac{i}{\hbar} [\hat{A}', \hat{H}] \rightarrow$ Heisenberg EOM.

Heisenberg Eq. of Motion

$\frac{d\langle \hat{A} \rangle}{dt} = -\frac{i}{\hbar} [\hat{H}, \hat{A}]$

Liouville-van Neuman Eq.

$\frac{d\rho}{dt} = \frac{i}{\hbar} [\hat{H}, \hat{\rho}]$

$\rho(t) = |\psi(t)\rangle \langle \psi(t)|$

$= \hat{U} |\psi(0)\rangle \langle \psi(0)| \hat{U}^\dagger$

$= \hat{U} \rho(0) \hat{U}^\dagger$

$\frac{d\rho}{dt} = \frac{d\hat{U}}{dt} \rho(0) \hat{U}^\dagger + \hat{U} \rho(0) \frac{d\hat{U}^\dagger}{dt}$

$= -\frac{i}{\hbar} \hat{U} \rho(0) \hat{U}^\dagger + \hat{U} \rho(0) \hat{U}^\dagger \left(\frac{i}{\hbar} \hat{H}\right)$

$= -\frac{i}{\hbar} \hat{H} \rho(t) + \frac{i}{\hbar} \rho(t) \hat{H}$

$\frac{d\rho(t)}{dt} = \frac{i}{\hbar} [\hat{H}, \rho(t)] \rightarrow$ Liouville von-Neumann eq.

$= \frac{1}{i\hbar} [\hat{H}, \rho(t)] \rightarrow$ Read Goldstein.

Phase space density $\sim$ density matrix.

For pure state: $\vec{\alpha} = \vec{n}$; $|\vec{\alpha}| = 1$.

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix}$$

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} \cos^2 \frac{\theta}{2} & e^{-i\phi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} & \sin^2 \frac{\theta}{2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1+\alpha_3}{2} & \frac{\alpha_1 - i\alpha_2}{2} \\ \frac{\alpha_1 + i\alpha_2}{2} & \frac{1-\alpha_3}{2} \end{pmatrix} = \begin{pmatrix} \frac{1+\langle\sigma_z\rangle}{2} & \frac{\langle\sigma_x\rangle - i\langle\sigma_y\rangle}{2} \\ \frac{\langle\sigma_x\rangle + i\langle\sigma_y\rangle}{2} & \frac{1-\langle\sigma_z\rangle}{2} \end{pmatrix}$$

Shrinking of Ball $\rightarrow$ Related to Decoherence.

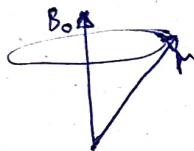
Question

1. Arbitrary gate $\rightarrow$ Rotation axis is arbitrary $\rightarrow$ Use Euler decomposition to describe it by known angle and find the gate

2. Liouville von Neuman equation:

$$\frac{d\langle A \rangle}{dt} = \text{Tr} \left(\frac{d\rho}{dt} H \right)$$

3.



$\frac{d\langle \mu \rangle}{dt}$ Zeeman Hamiltonian

$$H = g \mu_B \vec{\sigma} \cdot \vec{B} \quad B = B_0 \hat{k}$$

$$= g \mu_B B_0 \sigma_z$$

$$= \hbar \frac{\omega_0}{2} \sigma_z$$

$\Delta E = \hbar \omega_0$ between eigen modes

Zeeman field $\left\{ \begin{array}{l} \uparrow \\ \downarrow \end{array} \right\} \Delta E = \hbar \omega_0$

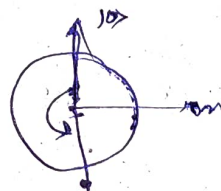
So for transition - we need σ_x or σ_y

So we have to place the frequency perpendicular to

Workout this ESR.

$$H = g \mu_B \underbrace{\vec{B} \cdot \vec{\sigma}}_{B_0 \hat{k}} + \gamma B_0 \cos(\omega t + \phi)$$

ESR



$$U = R_y(\pi)$$

$$= e^{i\frac{\pi}{2}} - i \sin \frac{\pi}{2} \sigma_x$$

$$= -i \sigma_x$$

Classical Information

Information content $\rightarrow$ lack of previous knowledge $\leftarrow$ gain knowledge ~~when~~ when told about an event.

111...1 $\rightarrow$ no information as it is predictable.

1010...1 $\rightarrow$ Random string $\rightarrow$ Gain maximum knowledge

~~0, 1, 2, 3, 4, 5, 6, 7~~ $\rightarrow$ 0, 1, 2, 3, 4, 5, 6, 7
8-possibility

Message Fixed Encoding

0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

(Why we are calling it 'encoding'?)

of bit required is 3.
 $2^3 = 8$

Surprise $\equiv$ Information gain

Suppose a quantity can assume one of m different possibilities then

$$2^n \geq m \text{ for some } n.$$

The smallest 'n' for which this holds tells us the number of bits is needed to represent or encode that quantity

$\Rightarrow$ Given a message 'm' how much info is contained in a given message?

Book by As $\rightarrow$ "Information Theory" $\rightarrow$ Maths.

Ralph Hartley (1929)

$m = 2^n$; $\log_2 m = n \leftrightarrow$ n bits can store in m diff. messages.

$m = 2^3 = 8$, $\log_2 8 = 3$; 3 bits are required to store 8 diff. messages.

For $m \leq 2^n \rightarrow$ min n bits are required.

$\triangleright$ A message that is unlikely to occur has a low probability and therefore a higher information contained

$\triangleright$ A message that is more likely to occur has a high probability and therefore a low information content contained.

Decrease uncertainty $\Rightarrow$ Decrease information
 Increase uncertainty $\rightarrow$ Increase info cont.

(Information is)
 Entropy
Book
 Extra Read:

There is some connection between information content and entropy.

Thus $I \propto \frac{1}{P}$ Probability.

2 Event 'A' and 'B' (Independent)
 $\downarrow \quad \downarrow$
 $\rightarrow$ Prob. $\rightarrow q \quad r$

Joint Probability $P(A \cap B) = P(A)P(B) = qr$

Then $I(A \cap B) \approx \frac{1}{q} \cdot \frac{1}{r} \neq$ not additive (1)

But $I = I_1 + I_2 = \frac{1}{q} + \frac{1}{r}$ (2)

(1) and (2) not consistent to each other.

Defn $\rightarrow I = -\log_2 P$ Because $\log_2(1/P) = -\log_2 P$.

Say, probability of Rain in Sahara Desert is $P_1 = 0.005$

$$I_1 = -\log_2 P_1 = 7.6439$$

Prob. of no rain in Sahara is $P_2 = 1 - P_1 = 0.995$.

$$I_2 = -\log_2 P_2 = 0.0072 \text{ (very less Information)}$$

III... 1 $\rightarrow$ Prob of occurrence of the bit '1' is again 1.

$$P_3 = 1$$

$$\text{So } I_3 = -\log_2 P_3 = 0$$

$\rightarrow$ Information functional (What is functional?)

Expectation value $\rightarrow \langle I \rangle = -\sum_i P_i \log_2 P_i = -\langle \log P \rangle$

0101... 1 $\rightarrow P(0) = 0.5$ $\sim$ Maximally mixed state
 couple Random $P(1) = 0.5$

$$\langle I \rangle = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = +\frac{1}{2} + \frac{1}{2} = 1 \text{ Highest}$$

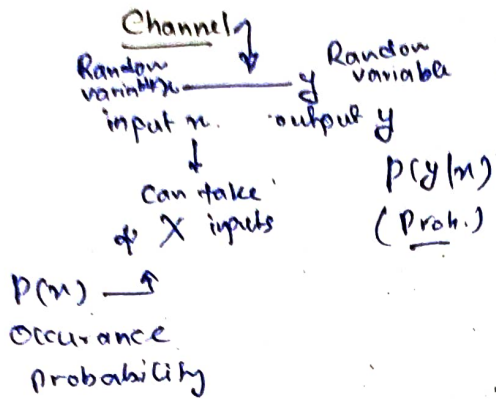
All possibilities are equally likely

Information content is highest for completely Random event.

Defn Information

States are not ~~restricted~~ restricted to Discrete state

Information theory quantifies the intuitive notion of surprise.
There are a set of events 'x', which can take values
 $\{x = 1, 2, \dots, X\} \rightarrow X \rightarrow$ possible values
for bits $X = 2$



$p(x)$ $\rightarrow$ Probability distribution for the ~~events~~ input message 'x' to appear at the input 'A'
 $P(y|x)$ $\rightarrow$ Probability for the channel to deliver a output y if the given input is x.

Let x_i be a R.V. which takes X possible values. Indexed by $i = 1, 2, \dots, X$.
Then the probability of seeing the i th value be p_i .

Definition: Information functional $H(p)$ $\rightarrow$ $H(p)$
which has the following properties.

(i) $H(p)$ is continuous $\rightarrow$ small increment in p will lead to small changes in $H(p)$

10/02/26

Lecture-13
Quantum Information Processing.

Property of $H(p)$

(i) $H(p)$ is continuous in 'p'

(ii) $H(p) \geq 0$ and $H(p) = 0$ iff just one p_i is non zero and all other $p_i = 0$. (ie. pure state)
For pure state Von Neuman Entropy is zero.

(iii) $H(p) \leq C(X)$, where $C(X)$ is a const that depends on the number of possible values of 'X'. (Upper Bound)

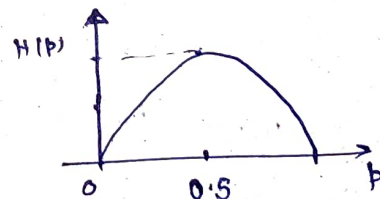
Suppose $X' > X \Rightarrow C(X') > C(X)$ The more the options are there, the less we know about the system.

Example:- (a) coin toss $p_{\text{head}} = \frac{1}{2}$ / $p_{\text{tail}} = \frac{1}{2}$

$$H(p) = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$$

$$= -p \log p - (1-p) \log (1-p)$$

$\rightarrow$ Max when $p = \frac{1}{2}$, $1-p = \frac{1}{2}$



(b) Dice $\rightarrow H(p) = -\frac{1}{6} \times 6 \times \log_2 \frac{1}{6} = +\log_2 6 = 2.585 + (1 + \log_2 3) > 1$
High info

Property - iv)

if 'x' is drawn from a distribution 'p' } all of independently drawn.
 if 'y' is drawn from a " 'q' } $P(A \& B) = P(A)P(B)$

$$\text{Then } H(p, q) = H(p) + H(q)$$

All those properties are satisfied by the Shanon entropy (Claude Shanon)

$$H(p) = - \sum_{i=1}^N p_i \log p_i$$

→ This is the definition of the entropy of a probability distribution.

(This definition is unique)

$$p_i = \frac{1}{N} \Rightarrow H(p) = - \frac{1}{N} \sum_{i=1}^N \log \frac{1}{N} = - \log \left(\frac{1}{N} \right) = \log N$$

(Remember, $S = k \log \Omega$)
 $N \rightarrow$ Number of accessible microstates

→ Prop (i) and prop (iv)

(Classical information (2) in NPTEL)

$$H = - \sum p_i \log p_i = - \langle \log p \rangle \rightarrow \text{Expectation value of } \log p$$

Consider a string $x_1, x_2, \dots, x_i, \dots, x_N$ of N samples. [Note that, for binary message $x_i \in \{0, 1\}$ otherwise it can take a max value X]

So we have X^N such message drawn from a distribution p and

let N_i be the number of times that the i th value of x_i occurs (or actually seen).

Because of the independence of the observation of the probability to see a particular string is the product of the individual probabilities.

$$p(x_1, x_2, \dots, x_N) = \prod_{i=1}^N p(x_i) \quad \text{--- (i)}$$

→ The product of the terms can be regrouped in terms of the possible values of x_i

$$p(x_1, x_2, \dots, x_N) = \prod_{i=1}^X p_i^{N_i} \quad \left(\begin{array}{l} X \rightarrow \text{distinct values of } x_i \\ \text{for a bit or qubit, } X=2 \\ N_i \Rightarrow \text{# of times occurs} \end{array} \right)$$

Taking log on both side and multiplying by $-1/N$,

$$-\frac{1}{N} \log p(x_1, x_2, \dots, x_N) = -\frac{1}{N} \log \left(\prod_{i=1}^X p_i^{N_i} \right) \approx - \sum_{i=1}^X \left(\frac{N_i}{N} \right) \log p_i$$

$$\approx - \sum_{i=1}^X p_i \log p_i \quad \left(\begin{array}{l} \text{Using law of large} \\ \text{numbers} \end{array} \right)$$

$$-\frac{1}{N} \log p(x_1, x_2, \dots, x_N) = - \sum_{i=1}^X p_i \log p_i = H(p) \quad \text{--- (iii)} \quad \left(\begin{array}{l} \text{as } N \rightarrow \infty \\ \left(\frac{N_i}{N} \rightarrow p_i \right) \end{array} \right)$$

Inverting it.

$$p(x_1, x_2, \dots, x_N) \approx 2^{-NH(x)} \quad \left(\text{assuming base-2} \right)$$

$$\text{OR } X^{-NH(x)} \quad \left(\text{general} \right)$$

Remarkable → The probability of seeing a particular string is independent of the elements of the string

⇒ Asymptotic Equipartition Property (AEP)

∴ The probability of occurrence for such a string is constant,
its inverse $\frac{1}{p} = 2^{NH(n)} \quad (x^{NH(n)})$

This gives the effective number of strings of that length. (instead of 2^N)

11/02/20

Lecture-14

$$H(n) = - \sum p_i \log_2 p_i = - \langle \log p_i \rangle \quad \& \quad p_n = 2^{-NH(n)} \quad \text{--- (iv)}$$

Remarkable → Does not depend on x & Probability of seeing a particular string
↳ The probability of seeing a particular string is independent of the element of the string (i.e. any x)

$$\frac{1}{p_x} = 2^{NH(n)} \rightarrow \text{Effective number of strings.}$$

(In binary choice)

Actual number of strings → larger.

$$x^N = 2^{N \log_2 x} \quad \text{--- (v)}$$

~~Is it x^N or 2^N~~

For bits → $x^N \Rightarrow 2^N \Rightarrow 2^{N \log_2 2}$

while we obtain $2^{NH(n)}$

This diff. is called makes data compression possible.

Shannon's Noiseless Coding Theorem

Shannon's Entropy ($H(n)$) reduces the effective number of bit used to conveying a message. For a string of length N we have 2^N possibility.

N bits → 2^N possible

↓ using a clever coding

$H(n)$ possibilities → $2^{H(n)}$ values

Instead of N bits we can use $H(n)$ bits by a clever coding.

$$H(n) < \log_2 x$$

Since, sample drawn from a distribution can on an avg be described by $H(n)$ bits rather than $\log_2 x$ bits, a coder can exploit this difference to store or transmit the string with $H(x)$ bits.

Example

(i) Source of message → information → message

4 symbols

0 00
1 01
2 10
3 11

→ without compression

$$\sum p_i = 1$$

The avg length of string is $H(p)$

$$H(p) = - \sum p_i \log_2 p_i = - \sum \frac{1}{4} \log_2 \frac{1}{4} = 2$$

distribution

#	Probability
1	$\frac{1}{2}$
2	$\frac{1}{4}$
3	$\frac{1}{8}$
4	$\frac{1}{8}$

$$\sum p_i = 1$$

Follow → Neilson
classical info.

Clever coding

Symbol	Prob.	Naive code	Clever code
1	$\frac{1}{2}$	00	0
2	$\frac{1}{4}$	01	10
3	$\frac{1}{8}$	10	110 or 100
4	$\frac{1}{8}$	11	111

The less prob → more bit.

$$-\frac{3}{4}$$

$$H(p) = -\sum p_i \log p_i = -\frac{1}{2} \log(\frac{1}{2}) - \frac{1}{4} \log(\frac{1}{4}) - 2 \times \frac{1}{8} \log(\frac{1}{8}) = \frac{7}{4} < 2$$

Avg. number of bits

Avg length of the message → $\bar{L} = \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \frac{1}{8} \times 3 + \frac{1}{8} \times 3$
 $= \frac{7}{4}$ (Same as Shannon Entropy)

Example - 2

Car Race

$C_1, C_2, \dots, C_8 \rightarrow 8$ cars.

The ~~winning~~ or winning probability →

P_1	C_1	$\frac{1}{2}$
P_2	C_2	$\frac{1}{4}$
P_3	C_3	$\frac{1}{8}$
P_4	C_4	$\frac{1}{16}$
P_5	C_5	$\frac{1}{64}$
P_6	C_6	$\frac{1}{64}$
P_7	C_7	$\frac{1}{64}$
P_8	C_8	$\frac{1}{64}$

$$\sum p_i = 1$$

$$H(p) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{4} \log \frac{1}{4} - \frac{1}{8} \log \frac{1}{8} - 4 \times (\log \frac{1}{64}) \times \frac{1}{64} = 2$$

Clever coding → less prob → high ~~code~~ bits

$$C_1 = 0$$

$$C_2 = 10$$

$$C_3 = 110$$

$$C_4 = 1110$$

$$C_5 = 111100$$

$$C_6 = 111101$$

$$C_7 = 111110$$

$$C_8 = 111111$$

Avg. length

$$\bar{L} = \sum p_i L_i$$

$$= 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{8}$$

$$+ 4 \times \frac{1}{16} + 4 \times 6 \times \frac{1}{64}$$

$$= 2$$

We gain 1 bit

1 bit less requ. to
~~show~~

Quantum Information:

$\rho \rightarrow \hat{\rho} \rightarrow$ density matrix.

$$H(\rho) = -\sum_i p(x_i) \log \{p(x_i)\} = -\langle \log \rho \rangle$$

↓
maximum for maximally mixed state (due to equal a priori probability)

Define, the measure of Quantum information

Von Neumann
Entropy

$$S = -\langle \log \rho \rangle = -\text{Tr}(\rho \log \rho)$$

↓ Density matrix

$$\rho_{\text{diag}} = U \rho U^\dagger = \rho \rho^{-1}, \quad P = (| | |) \quad \text{But } D = U^\dagger \rho U \quad \text{Note}$$

↓ Eigen vectors of ρ
 $P P^{-1} = \mathbb{I} = P^{-1} P$

$$S = -\text{Tr}(\underbrace{\rho \rho^{-1}}^{\mathbb{I}} \underbrace{\rho \log \rho \rho^{-1}}_{\rho \log \rho \rho^{-1}})$$

$$= -\text{Tr}(\underbrace{\rho}_{\rho_d} \log \rho \rho^{-1})$$

$$S = -\text{Tr}(P^{-1} \rho P P^{-1} \rho \log \rho)$$

$$= \text{Tr}(\rho \rho P^{-1} \rho \log \rho P^{-1}) \quad \text{cyclicity of trace}$$

Let there be a general matrix A :

$$A_d = P A P^{-1} \Rightarrow P^{-1} A_d P = A$$

$$e^A = \mathbb{I} + A + \frac{A^2}{2!} + \dots = \mathbb{I} + P^{-1} A_d P + \frac{P^{-1} A_d^2 P}{2!} + \dots$$

$$= P^{-1} (\mathbb{I} + A_d + \frac{A_d^2}{2!} + \dots) P$$

$$= P^{-1} e^{A_d} P$$

★ Note that
in case of
time evolution
 $\rho(t) = U \rho(0) U^\dagger$
But in case of
Diagonal matrix
 $\rho_{\text{diag}} = U^\dagger \rho U$

$$S, e^A \Rightarrow \boxed{P e^A P^{-1} = e^{A_d}}$$

$$A_d = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_N \end{pmatrix}$$

$$e^{A_d} = \begin{pmatrix} e^{\lambda_1} & & 0 \\ & e^{\lambda_2} & \\ 0 & & e^{\lambda_N} \end{pmatrix}$$

Michel
Artine algebra

Jordan Canonical form

$$f(A) \longrightarrow f(A_d) = P f(A) P^{-1} \quad (\text{Functional})$$

Prove skipped.
↓
check group.

$$\boxed{P \log \rho P^{-1} = \log(\rho_d)}$$

$$S = -\text{Tr}(\rho_d \log \rho_d) = -\text{Tr} \left[\begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_N \end{pmatrix} \begin{pmatrix} \log \lambda_1 & & 0 \\ & \log \lambda_2 & \\ 0 & & \log \lambda_N \end{pmatrix} \right]$$

$$= -\text{Tr} \left[\begin{pmatrix} \lambda_1 \log \lambda_1 & & 0 \\ & \lambda_2 \log \lambda_2 & \\ 0 & & \lambda_N \log \lambda_N \end{pmatrix} \right]$$

$$\boxed{S = -\sum_i \lambda_i \log \lambda_i} \rightarrow \text{Von Neumann equation.}$$

$$\underline{H(p(x)) = -\sum_i p_i \log p_i}$$

For Pure state $\rightarrow \lambda_1 = 1$ & $\lambda_{i \neq 1} = 0$

$$S(\rho) = -1 \log_2 1 - 0 \log_2 0 = 0$$

~~slowly varying~~ $\rightarrow$ slow varying function
so $0 \log_2 0 = 0$

$S(\rho) = 0$ for pure state.

12/02/26

Lecture-15

Von-Neumann Entropy

$$S(\rho) = -\text{Tr}(\rho \log \rho) = -\sum \lambda_i \log \lambda_i$$

$$|4\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

$$\rho = \begin{pmatrix} \cos^2 \frac{\theta}{2} & \frac{e^{-i\varphi} \sin \theta}{2} \\ \frac{e^{i\varphi} \sin \theta}{2} & \sin^2 \frac{\theta}{2} \end{pmatrix}$$

$$\lambda_1 = 1, \lambda_0 = 0$$

$$S(\rho) = -1 \log 1 - 0 \log 0 = 0$$

$\rightarrow \log$ is a slowly varying fn than x

Mixed state $\rightarrow$

$$\rho = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix} = 1/4 \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

$$S(\rho) = -\frac{3}{4} \log \frac{3}{4} - \frac{1}{4} \log \frac{1}{4}$$

$$= -\frac{3}{4} + \frac{1}{4} (-3 \log 3 + 3 \log 4 + \log 4)$$

3.26

$$= \frac{1}{4} (-3 \times 1.58 + 3 \times 2 + 2)$$

$$= \frac{1}{4} \times (-4.74 + 8)$$

$$= \frac{3.26}{4} = 0.82 \neq 0$$

$$\rho = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} \quad S(\rho) = -1/2 \log(1/2) - 1/2 \log(1/2) = 1$$

For a maximally mixed state $S(\rho) = 1 \Rightarrow$ Maximal value of Von-Neumann Entropy.

• Entanglement $\rightarrow$ Must be ≥ 2 qubit system

$\mathcal{H}_A \otimes \mathcal{H}_B \rightarrow 4$ dim Hilbert space
 $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

Reminder

$$|00\rangle = |0\rangle \otimes |0\rangle$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$|01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}; |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}; |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\rho = |00\rangle\langle 00|$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} (1000)$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$S(\rho) = 0$$

So only one eigen value is one

$$\rho_2 = |01\rangle\langle 01| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \text{ diag}(0, 1, 0, 0) \quad \rho_3 = |10\rangle\langle 10| = \text{diag}(0, 0, 1, 0)$$

$$\rho_4 = |11\rangle\langle 11| = \text{diag}(0, 0, 0, 1)$$

For all the case Von Neumann Entropy is zero

~~Separable~~

~~Separable state~~ Seperable State

$$|\psi\rangle_A \in \mathcal{H}_A, |\phi\rangle_B \in \mathcal{H}_B$$

$|\chi\rangle_{AB} = |\psi\rangle_A \otimes |\phi\rangle_B$ is written in the form of direct product state, and is termed as "seperable state"

1st qubit ~~belongs~~ belongs to $\mathcal{H}_A$
2nd qubit belongs to $\mathcal{H}_B$

Eg.

$$\begin{aligned} |\psi\rangle_{AB} &= \frac{1}{2} (|00\rangle_{AB} + |01\rangle_{AB} + |10\rangle_{AB} + |11\rangle_{AB}) \\ &= \frac{1}{2} (|0\rangle_A \otimes (|0\rangle + |1\rangle)_B + |1\rangle_A \otimes (|0\rangle + |1\rangle)_B) \\ &= \frac{1}{2} ((|0\rangle_A + |1\rangle_A) \otimes (|0\rangle_B + |1\rangle_B)) \\ &= \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)_A \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)_B \\ &= |+\rangle_A \otimes |+\rangle_B \rightarrow |\psi\rangle_A \otimes |\phi\rangle_B \rightarrow \text{seperable state.} \end{aligned}$$

$$\begin{aligned} |\psi\rangle &= \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle) \\ &= \frac{1}{2} (|0\rangle \otimes (|0\rangle - |1\rangle) + |1\rangle \otimes (|0\rangle - |1\rangle)) \\ &= \frac{1}{2} \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= |+\rangle \otimes |-\rangle \rightarrow \text{seperable state.} \end{aligned}$$

$$\begin{aligned} |\psi\rangle_{AB} &= \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle + |11\rangle) \\ &= \frac{1}{2} (|0\rangle (|0\rangle - |1\rangle) + |1\rangle (|0\rangle + |1\rangle)) \end{aligned}$$

Can't write as a Seperable states of individual vectors in $\mathcal{H}_A$ and $\mathcal{H}_B$.

→ Can't

This is an Entangled state.

Entangled state: If a bipartite state can't be written as a direct product of its constituent parties then the composite states is called Entangled state.

Most Famous Bell's state examples →

Bell (John Bell) states

$$(i) |\psi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$(ii) |\psi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$(iii) |\phi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

$$(iv) |\phi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \rightarrow \text{singlet states.}$$

~~$\rho(1\psi^-)$~~

$$|\psi^+\rangle = \frac{1}{\sqrt{2}} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$|\psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} ; |\psi^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\rho(|\psi^-\rangle) = |\psi^-\rangle \langle \psi^-|$$

$$= \frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \rho^2 = \rho$$

It is pure state by construction.

0 0 0 0

$$\det \begin{vmatrix} -\lambda & 0 & 0 & 0 \\ 0 & \frac{1}{2} - \lambda & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} - \lambda & 0 \\ 0 & 0 & 0 & -\lambda \end{vmatrix} \Rightarrow -\lambda \begin{vmatrix} \frac{1}{2} - \lambda & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} - \lambda \end{vmatrix} = -\lambda \left(\left(\frac{1}{2} - \lambda \right)^2 - \frac{1}{4} \right) = -\lambda \left(\lambda^2 - \lambda \right) = 0$$

$$\lambda^2 (\lambda - 1) = 0$$

$$\lambda = 0, \lambda = 0, \lambda = 1, \lambda = 0$$

$S(\rho(|\psi^-\rangle)) = 0 \rightarrow$ Von Neumann entropy is zero.

Reduced Density Matrix

$\rho_{AB} \rightarrow 2 \times 2$ matrix

$\downarrow$
4x4

$\rho_A \rightarrow$ Defined as the reduced matrix obtained by tracing out the coordinates or degrees of freedom of the party 'B'.
The phenomenon is called partial tracing.

$$\text{defn } \rho_A = \mathbb{I}_A \otimes \sum_B \langle 0| \rho_{AB} | \mathbb{I}_A \otimes |0\rangle_B + \mathbb{I}_A \otimes \sum_B \langle 1| \rho_{AB} | \mathbb{I}_A \otimes |1\rangle_B$$

$$\rho_{AB} = |\psi^-\rangle \langle \psi^-|$$

$$= \frac{1}{2} (|01\rangle - |10\rangle) \frac{1}{\sqrt{2}} (\langle 01| - \langle 10|)$$

$$= \frac{1}{2} (|01\rangle \langle 01| - |10\rangle \langle 01| - |01\rangle \langle 10| + |10\rangle \langle 10|)$$

$$\rho_A = \frac{1}{2} \mathbb{I}_A$$

$$\rho_A = \mathbb{I}_A \otimes \sum_B \langle 0| \frac{1}{2} (|01\rangle \langle 01| - |10\rangle \langle 01| - |01\rangle \langle 10| + |10\rangle \langle 10|) | \mathbb{I}_A \otimes |0\rangle_B$$

$$+ \mathbb{I}_A \otimes \sum_B \langle 1| \frac{1}{2} (|01\rangle \langle 01| - |10\rangle \langle 01| - |01\rangle \langle 10| + |10\rangle \langle 10|) | \mathbb{I}_A \otimes |1\rangle_B$$

check calculation $\rightarrow$ in first term

only last term contributes.

(b) in second term

only first term contribute

$$\langle 0| \frac{1}{2} (|01\rangle \langle 01| - |10\rangle \langle 01| - |01\rangle \langle 10| + |10\rangle \langle 10|) | \mathbb{I}_A \otimes |0\rangle_B$$

$$\rho_A = \frac{1}{2} \left(\underset{\substack{\uparrow \\ \text{from first} \\ \text{term}}}{|1\rangle\langle 1|} + \underset{\substack{\uparrow \\ \text{from} \\ \text{second term}}}{|0\rangle\langle 0|} \right) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{Maximally mixed state}$$

$$S(\rho_A) = -\frac{1}{2} \log\left(\frac{1}{2}\right) \times 2 = 1 \rightarrow \text{Maximally entangled state}$$

★ If is the signature of entangled state $\rightarrow$ Normal density matrix has zero Entropy while the reduced one have entropy.

13/02/26

Lecture-16

Ex Entanglement

$$\rho_{AB} = |00\rangle_{AB} \langle 00|_{AB} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow S(\rho) = 0$$

$$\text{Reduced density matrix } \rho_A = \mathbb{I}_A \otimes \langle 0|_B \rho_{AB} | \mathbb{I}_A \otimes |0\rangle_B + \mathbb{I}_A \otimes \langle 1|_B \rho_{AB} | \mathbb{I}_A \otimes |1\rangle_B$$

$$\begin{aligned} \rho_A &= \mathbb{I}_A \otimes \langle 0|_B |00\rangle \langle 00| \mathbb{I}_A \otimes |0\rangle_B + \mathbb{I}_A \otimes \langle 1|_B |00\rangle \langle 00| \mathbb{I}_A \otimes |1\rangle_B \\ &= |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \rightarrow \text{Also a pure state} \end{aligned}$$

$$S(\rho_A) = 0 \rightarrow \text{separable state}$$

$$\rho_{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}_A \otimes \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}_B$$

But for entangled state we can't write in terms of the direct product.

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\text{Ex } |\psi\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$= \frac{1}{2} (|0\rangle + |1\rangle)_A \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)_B = |+\rangle_A \otimes |+\rangle_B$$

$$\rho_{AB} = (|+\rangle_A \otimes |+\rangle_B) (\langle +|_A \otimes \langle +|_B)$$

$$= \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}_A \otimes \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}_B$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \otimes \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \rightarrow \text{Can be written in terms of multiplication}$$

$$|\psi\rangle_{AB} = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

$$\rho_{AB} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$S(\rho_{AB}) = 0$$

$$\lambda_1 = 1, \lambda_2 = \lambda_3 = \lambda_4 = 0$$

$$\text{But, } \rho_A = \text{Tr}_B(\rho_{AB}) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \underline{S(\rho_A) = 1} \quad (\text{Maximally mixed state})$$

$$\rho_A = \text{Tr}_B(\rho_{AB})$$

$$= \mathbb{I}_A \otimes \sum_B \rho_{AB} |I_A \otimes 10\rangle_B + \mathbb{I}_A \otimes \sum_B \rho_{AB} |I_A \otimes 11\rangle_B$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes (10) \begin{pmatrix} \rho_{AB} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes (01) \begin{pmatrix} \rho_{AB} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$+ \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 0 & -1 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$|\psi(t)\rangle = e^{-i\omega t/2} \otimes |0\rangle \\ = \begin{pmatrix} \cos \frac{\omega t}{2} & -i \sin \frac{\omega t}{2} \\ -i \sin \frac{\omega t}{2} & \cos \frac{\omega t}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} \cos \frac{\omega t}{2} \\ -i \sin \frac{\omega t}{2} \end{pmatrix} \xrightarrow{\frac{\omega t}{2}} \text{compare with the state}$$

Partial Self Doing Reduced Density matrix for other Bells state

⚡ Bell States

$$(i) |\psi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad (ii) |\psi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$(iii) |\psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) \quad (iv) |\psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

$$(i) \rho_{AB} |\psi^+\rangle = |\psi^+\rangle \langle \psi^+| = \frac{1}{4} (|00\rangle \langle 00| + |11\rangle \langle 11| + |00\rangle \langle 11| + |11\rangle \langle 00|)$$

$$= \frac{1}{2} (|00\rangle \langle 00| + |11\rangle \langle 11|)$$

$$= \frac{1}{2} (|00\rangle \langle 00| + |11\rangle \langle 11|)$$

$$\rho_A = \mathbb{I}_A \otimes \langle 0| \rho_{AB} |\mathbb{I}_A \otimes |0\rangle_B + \mathbb{I}_A \otimes \langle 1| \rho_{AB} |\mathbb{I}_A \otimes |1\rangle_B$$

$$= \mathbb{I}_A \otimes \langle 0| \frac{1}{2} (|00\rangle \langle 00| + |11\rangle \langle 11| + |00\rangle \langle 11| + |11\rangle \langle 00|) \mathbb{I}_A \otimes |0\rangle_B$$

$$+ \mathbb{I}_A \otimes \langle 1| \frac{1}{2} (|00\rangle \langle 00| + |11\rangle \langle 11| + |00\rangle \langle 11| + |11\rangle \langle 00|) \mathbb{I}_A \otimes |1\rangle_B$$

$$= \mathbb{I}_A \otimes \frac{1}{2} (|0\rangle \langle 0| + |1\rangle \langle 1|) = \frac{1}{2} \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right)$$

$$= \frac{1}{2} \left[\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \text{Maximally mixed state.}$$

$$(ii) |\psi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$\rho_{AB} |\psi^-\rangle = |\psi^-\rangle \langle \psi^-|$$

$$= \frac{1}{2} (|00\rangle - |11\rangle) (\langle 00| - \langle 11|)$$

$$= \frac{1}{2} (|00\rangle \langle 00| - |11\rangle \langle 00| - |00\rangle \langle 11| + |11\rangle \langle 11|)$$

$$\rho_A = \mathbb{I}_A \otimes \langle 0| \rho_{AB} |\mathbb{I}_A \otimes |0\rangle_B + \mathbb{I}_A \otimes \langle 1| \rho_{AB} |\mathbb{I}_A \otimes |1\rangle_B$$

$$= \frac{1}{2} (|0\rangle \langle 0| + |1\rangle \langle 1|) = \frac{1}{2} \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right)$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \text{Maximally mixed state}$$

100>

(iii) ✓

$$(iv) |\psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

$$\rho_{AB} = |\psi^+\rangle \langle \psi^+| = \frac{1}{2} (|01\rangle + |10\rangle) (\langle 01| + \langle 10|)$$

$$= \frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\rho_A = \text{Tr}_B(\rho_{AB})$$

$$= \mathbb{I}_A \otimes \langle 0| \rho_{AB} |\mathbb{I}_A \otimes |0\rangle_B + \mathbb{I}_A \otimes \langle 1| \rho_{AB} |\mathbb{I}_A \otimes |1\rangle_B$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rho_{AB} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rho_{AB} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \frac{1}{2}$$

$$\Rightarrow \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \text{Maximally mixed state}$$

$$\alpha^2, \beta^2, \alpha + \beta$$

$$2(\alpha + \beta)$$

$$\frac{\alpha^2}{2} + \frac{\beta^2}{2}$$